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Environment Mapping with 28 GHz Beam Steering Transcei- vers

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Abstract

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Millimeter wave communications are central to 5G New Radio and require directional transmission and reception to overcome severe path loss. This introduces the need for efficient beam alignment during the initial access procedure, where the status of suitable search algorithms remains an open question in the standardization process. The objective of this thesis is to investigate, implement, and eventually experimentally evaluate candidate strategies for initial beam detection, with a particular focus on feasibility under standard-compliant constraints and practical hardware limitations.

The methodology combines a thorough review of state-of-the-art approaches with a selective implementation of representative search strategies. Exhaustive search, hierarchical refinement, among others are tested using a complete over-the-air setup involving commercial mmWave transceiver modules. The experiments address both favorable line-of-sight and challenging non-line-of-sight scenarios, enabling an assessment of algorithm robustness under non trivial conditions.

The results show that exhaustive scans, although theoretically offering the finest angular resolution, demand excessive signaling and time resources that make them costly for real 5G deployments. In contrast, hierarchical search strategies consistently converge toward comparable beam pairs solution, delivering strong received signal quality and stable SNR values despite the reduced angular precision of wide beams in the early stages. The refinement phases effectively compensate for potential initial misdetections of the dominant path, highlighting the robustness of this approach. Ultimately, angular deviations, defined as the error between the theoretical beam pair angle combination and the estimated propagation path, are inherently constrained by the finite beamwidth of the hardware, even when dense angular sampling is employed.

In conclusion, this work demonstrates that low-complexity search strategies represent a practical and efficient compromise for initial access in mmWave 5G systems. They combine feasibility, robustness, and compliance with standardization constraints, while offering performance sufficiently close to exhaustive search to justify their adoption in real-world deployments.

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1 Introduction

1.1 Evolution of Mobile Generations

With the ever-increasing demand for higher data rates, heavier traffic loads, the proliferation of connected smart devices, and the resulting high-bandwidth applications, there is a important need for network infrastructures capable of delivering reliable services with specific performance characteristics. Modern applications leveraging such networks typically require ultra-high data rates, low latency, high reliability, mobility support, flexibility, scalability, and massive connectivity.

These capabilities underpin the next generation of technologies, enabling use cases such as real-time traffic monitoring in smart cities, vehicle-to-everything (V2X) communications, augmented reality, ultra-low-latency cloud gaming, and large Internet of Things (IoT) deployments, as depicted in figure [1.1].

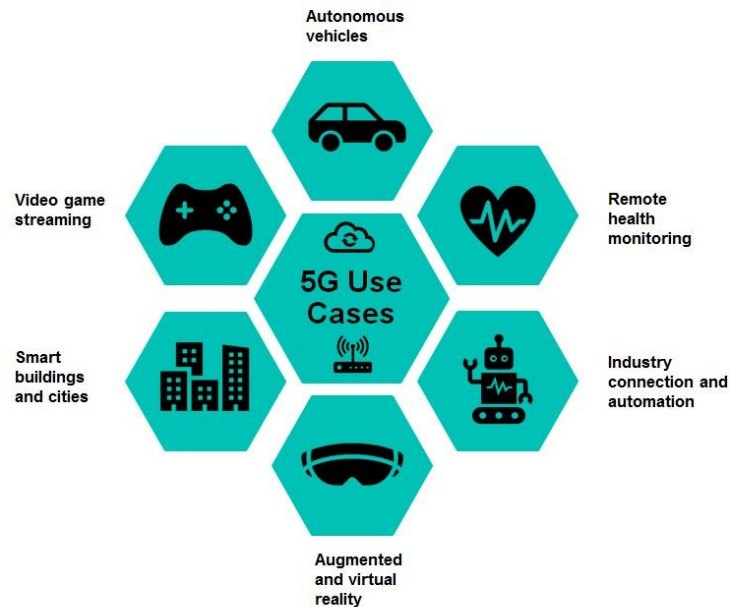


FIGURE 1.1 – Modern applications use cases. [9]

Historically, the first generation of mobile communications, introduced around 1980, was based on analog transmission and primarily enabled voice telephony, with the Advanced Mobile Phone System (AMPS) being the dominant technology [30]. This system

laid the foundations of cellular communication [3], delivering analog voice services using frequency modulation in the 800-900 MHz range. However, due to the inherent limitations of analog transmission, signal quality was often degraded by significant noise and interferences [44].

The second generation, emerging around 1990, transitioned to digital transmission over the radio link [59]. This enabled basic digital signal processing, resulting in improved voice quality, greater spectral efficiency, and encryption capabilities, in addition to the introduction of very limited data services. The Global System for Mobile Communications (GSM) became the dominant standard during this era [30][44].

The third generation, deployed in the early 2000s, marked the introduction of multimedia services and higher data rates in addition to voice. It offered a significant improvement in mobile broadband quality and enabled fast wireless Internet access for end users [22].

More recently, the fourth generation, represented by Long Term Evolution (LTE) technology, provided a unified architecture with higher efficiency and substantially greater end-user data rates. This was achieved through orthogonal frequency-division multiplexing (OFDM)-based transmission schemes, enabling the use of wider bandwidths. Generations from 1G to 4G have predominantly operated in the ultra-high frequency (UHF) and lower microwave regions of the electromagnetic spectrum [58].

The fifth generation introduces a fundamental shift with the formal inclusion of higher frequency bands, notably in the millimeter wave region of the spectrum. The 5G New Radio (NR) standard is typically deployed in bands around 26 GHz, 28 GHz, and 37-39 GHz, among others [56].

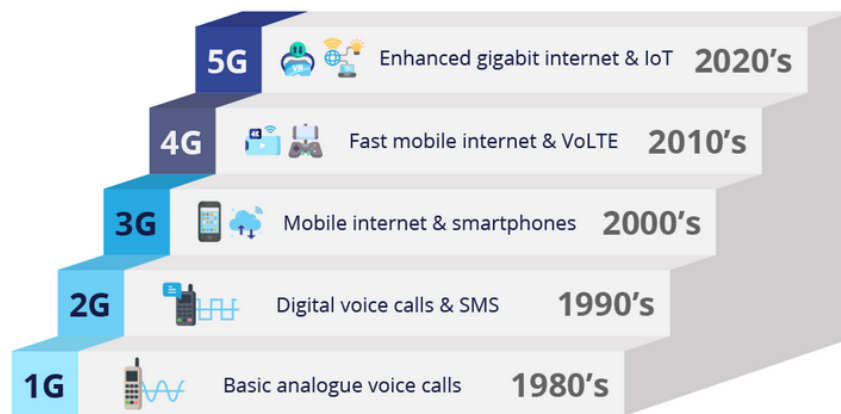


FIGURE 1.2 – Evolution from 1G to 5G [30]

In summary, the fifth generation of mobile communication systems marks a significant improvement, aiming to fulfill the requirements mentioned above with the extension of the operational spectrum beyond traditional sub-6 GHz bands.

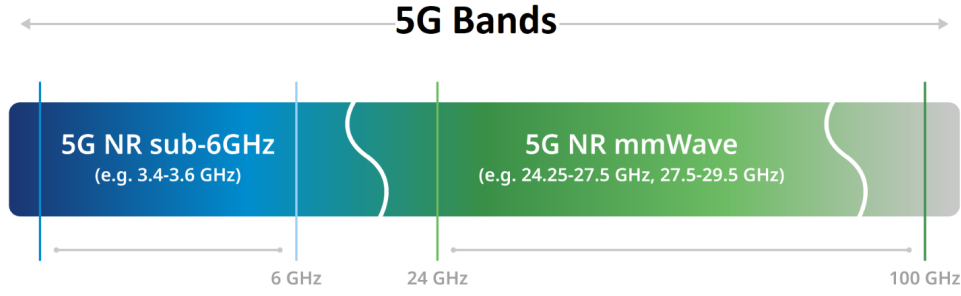


FIGURE 1.3 – Addition of mmWave bands in the 5G standard [64].

1.2 Millimeter Wave Communications

Millimeter wave (mmWave) refers to the portion of the electromagnetic spectrum spanning approximately 30 GHz to 300 GHz, corresponding to wavelengths between 1 mm and 10 mm. The adoption of mmWave frequencies in current and future generations of wireless communication systems, is of particular interest due to their physical and spectral characteristics. In particular, the available bands offer substantially wider channel bandwidths, typically on the order of several hundred megahertz, directly addressing the increasing demand for ultra-high-speed and high-capacity wireless services.

From a hardware perspective, the shorter wavelengths associated with mmWave frequencies enable the integration of compact, high-density antenna arrays into user equipment (UE). This is an important enabler for advanced beamforming and *Multiple Input Multiple Output* (MIMO) techniques, which are essential to counteract the severe attenuation inherent to mmWave propagation and to maintain robust links. Moreover, mmWave operation in 5G NR is inherently beam-centric, as the standard relies on highly directional transmissions.¹

Although beyond the direct scope of this work, it is noteworthy that mmWave frequencies are also well suited for sensing and localization applications, due to their large bandwidth and high spatial resolution. These features are particularly relevant in emerging joint communication and sensing paradigms envisioned for 6G systems. Narrow beam architectures provide high angular precision, while wide bandwidths facilitate fine range resolution [36]². The relevance of millimeter waves is thus expected to grow in the upcoming generations as well.

1.2.1 Challenges and constraints

Millimeter wave technology presents several notable challenges and limitations. A first constraint is the significantly higher path loss compared to lower-frequency bands.

1. Section 1.2.2 further explains the concept of *Directional Transmission*.

2. Authors in [36] defines range resolution as "*the ability to distinguish between two or more objects*" in the context of Radar applications. It is demonstrated that Range Resolution is proportional to the achievable bandwidth, as mentioned here.

Equation 1.1 presents the Friis formula for free-space propagation, where P_t is the transmitted power, G_t and G_r denote the transmitter and receiver antenna gains, λ is the wavelength, and d is the distance between the devices. This relationship makes it clear that, for a given transmission distance and antenna configuration, the received power decreases quadratically with frequency.

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 \quad (1.1)$$

Another major challenge is the strong susceptibility of mmWave signals to environmental obstructions. These signals experience high penetration losses through common building materials [17]. Furthermore, the relative size of obstacles with respect to the wavelength induces that mmWave signals exhibit poor diffraction and limited scattering, which further constrains the ability to propagate around obstructions. As a result, maintaining connectivity often requires a clear line-of-sight (LoS) or a strong reflected path.

These propagation limitations lead to smaller cell sizes and restricted coverage ranges for base stations (BS). Consequently, dense and spatially distributed deployments of access points are necessary in urban and indoor environments to ensure a decent link quality. While operating in overlapping frequency bands within a cell can introduce intra-cell interference, this risk is greatly reduced by the highly directional nature of mmWave transmissions³ [48]. This directionality enables efficient frequency reuse between relatively close users, avoiding interference issues encountered in omnidirectional systems at lower frequencies.

From a hardware perspective, UE faces several constraints when supporting mmWave operation. Operating over wide bandwidths requires high-speed analog-to-digital (ADC) and digital-to-analog (DAC) converters, and wideband filters, all of which increase circuit complexity and power consumption [49]. Moreover, enabling advanced MIMO techniques such as fully digital beamforming necessitates multiple independent RF chains, each with its own ADC/DAC and front-end module, further increasing power consumption. In contrast, analog beamforming, and to some extent hybrid beamforming, shares RF chain components across antennas, thereby reducing hardware complexity and power requirements.

1.2.2 Beamforming

The concepts of *directional transmission* and *MIMO*, mentioned in the previous section, are closely linked to *beamforming*, a key technique for mitigating the high attenuation illustrated by Equation 1.1.

Beamforming is a signal processing method applied in antenna arrays to direct the transmission or reception of wireless signals toward specific angular directions, rather than radiating or receiving energy uniformly in all directions. The principle relies on the constructive and destructive interference of signals emitted from multiple antenna elements arranged in a structured array. By precisely adjusting the phase and amplitude of the signal at each antenna element, the radiated electromagnetic waves can be made to interfere constructively in the desired direction and destructively elsewhere.

3. See Section 1.2.2.

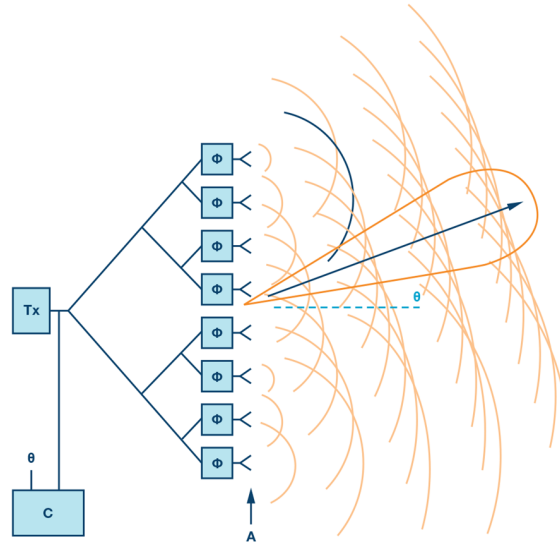


FIGURE 1.4 – Simultaneous transmissions from multiple antennas form a spatially confined beam [46].

The directional gain achieved through beamforming does not result from increased radiated power, but from the spatial concentration of energy that would otherwise be spread. This produces a confined main lobe of radiation, referred to as a beam, with significantly higher signal amplitude in the targeted direction.

Increasing the number of antenna elements enables the formation of narrower beams with finer angular resolution, a property that underpins the inherently directional nature of mmWave transmissions.

A fundamental distinction exists between analog and digital beamforming architectures, each defined by the signal processing domain in which the beam is shaped and the corresponding hardware requirements.

In analog beamforming, the beam pattern is formed by applying phase shifts to the signal at each antenna element directly in the analog domain, typically through a network of tunable phase shifters, as shown in Figure 1.5. All elements share a single RF chain, which restricts operation to one beam at a time. Consequently, when sweeping through different directions, the beam must be updated sequentially. This inherent limitation makes analog beamforming less suitable for scenarios requiring simultaneous multi-directional transmissions, as illustrated in Figure [1.7].

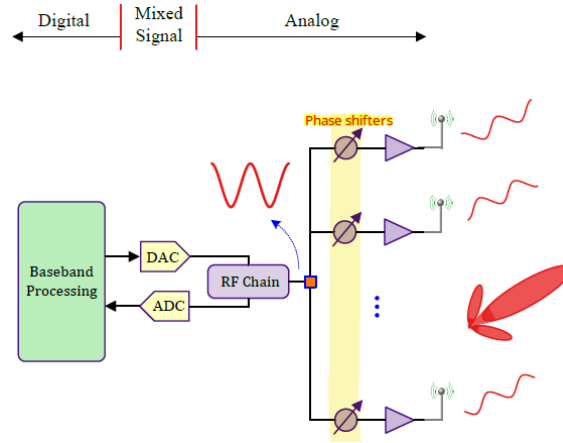


FIGURE 1.5 – Analog beamforming implemented through controlled phase shifters [20].

In contrast, digital beamforming operates in the baseband domain, with each antenna element connected to its own RF chain and dedicated front-end. This configuration allows precise, independent control of the weights applied to each element, enabling for example the creation of multiple simultaneous beams. Such flexibility supports advanced techniques like as multi-user MIMO, in which different beams are directed toward different users concurrently.

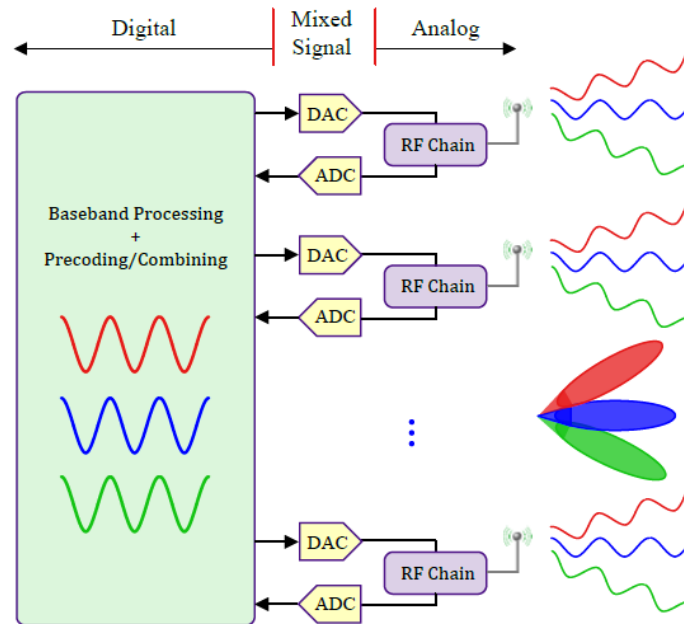


FIGURE 1.6 – Example of digital beamforming, enabling multiple spatial streams in parallel [20].

Given these trade-offs, analog beamforming or hybrid beamforming⁴ is often preferred in practical mmWave systems, particularly in user equipment. Analog beam steering

4. Hybrid beamforming combines a reduced number of RF chains with analog phase control, balancing flexibility with hardware complexity.

specifically provides a cost-effective mean to dynamically adjust the beam direction between a user terminal and a base station. Thereby mitigating environmental blockages and severe path loss without incurring the hardware overhead of fully digital architectures.

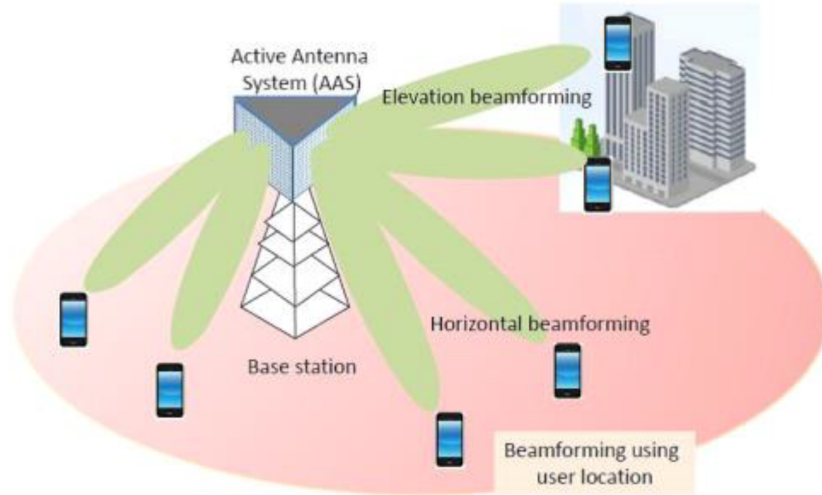


FIGURE 1.7 – Multiple users example highlighting potential need for concurrent streaming scenarios [15].

1.3 Objective of the Thesis

The set of operations involved in dynamically adapting beams at both the transmitter (Tx) and receiver (Rx) is collectively referred to as *beam management*. Given the inherently beam-centric nature of millimeter wave systems, these procedures are essential for ensuring reliable directional communication.

Among these operations, one of the most critical is the initial access phase, during which the UE and the base station must determine an optimal transmit-receive beam pair to establish a connection. This process requires a spatial search across multiple beam combinations which can introduce considerable overhead in both time and energy, especially with exhaustive solutions. The development of efficient, low overhead, and standard-compliant approaches to address this challenge has therefore become a prominent topic in academic research.

In this context, the objective of this master thesis is to investigate and evaluate a range of algorithmic approaches for initial beam alignment in millimeter wave communication systems. This work focuses specifically on the initial beam detection process between two devices, with particular emphasis on practical implementation and adherence to 5G standardization constraints. By experimentally reproducing initial access algorithms under realistic hardware limitations, this study aims to provide insights into their relative performance, implementation feasibility, and overall suitability for 5G communications.

2 The 5G *New Radio* Standard

As initial beam alignment algorithms are strongly influenced by the protocol constraints, a clear understanding of the 5G standard is essential. The standard specifies the timing, resource allocation, and beam management mechanisms that ultimately determine the feasibility of beam-based algorithms.

The fifth generation of mobile communication systems has been designed to support flexible, high-throughput, and low-latency communications. It classifies services into three primary use-case categories, each associated with a set of formalized performance requirements.

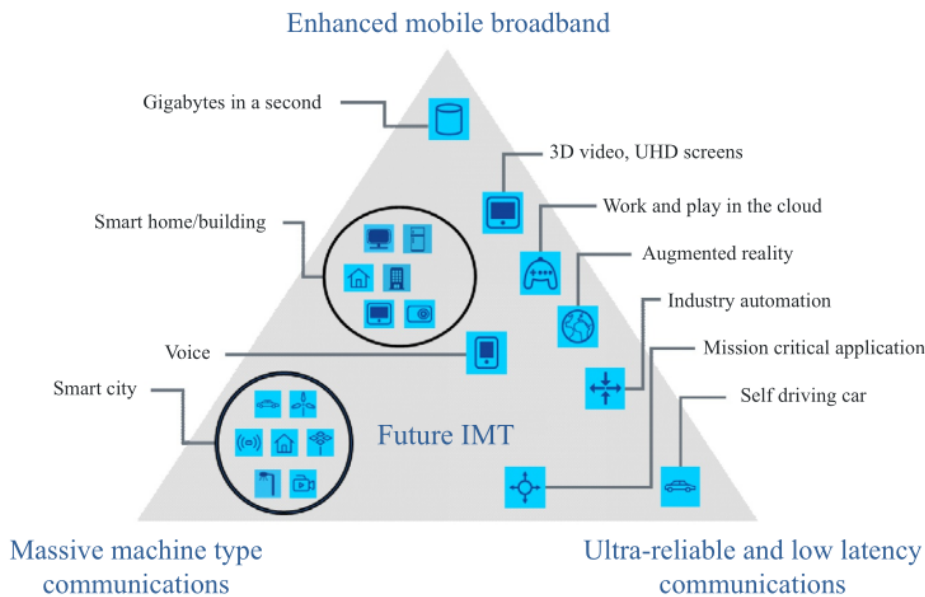


FIGURE 2.1 – 5G use cases and their mapping to usage scenarios [23].

2.1 Resource Allocation in the Time–Frequency Domain

The 5G frame structure follows a time-domain hierarchy reminiscent of LTE but with greater flexibility. A radio frame has a fixed duration of 10ms and is subdivided into ten 1ms subframes. Each subframe contains multiple slots whose duration depends on the numerology (defined below). A slot consists of a sequence of Orthogonal Frequency-Division Multiplexing (OFDM) symbols. With normal cyclic prefix, 14 symbols per slot

are used across numerologies. A key enhancement over LTE is the introduction of mini-slots, which allow transmissions to start at any OFDM symbol boundary and occupy only a subset of symbols within a slot. This design enables low-latency, fine-grained scheduling and is well suited to time sensitive services such as Ultra-Reliable Low-Latency Communications (URLLC).

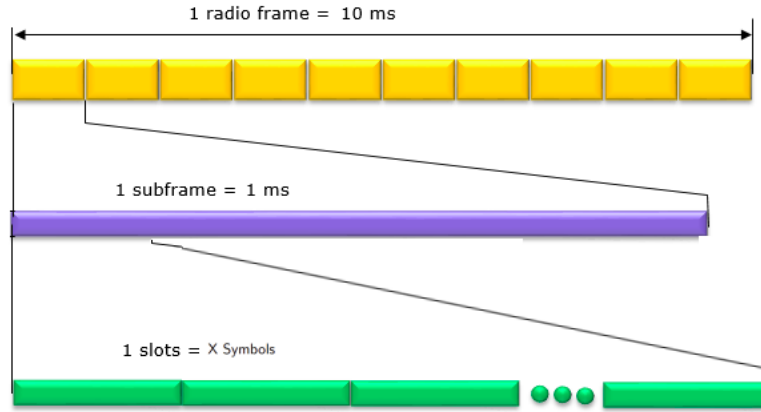


FIGURE 2.2 – Basic NR frame layout [7].

In the frequency domain, the allocated channel bandwidth is partitioned into a set of equally spaced subcarriers. The separation between adjacent subcarriers, referred to as the Subcarrier Spacing (SCS), controls the spectral sampling of the allocated band and thus aligns naturally with the discretization of the electromagnetic spectrum into frequency bins.

Because symbol duration is inversely proportional to subcarrier spacing¹, a larger SCS yields shorter OFDM symbols, whereas a smaller SCS yields longer ones. This frequency-domain partitioning ties directly to OFDM, which provides the bridge between the time-domain framing and the frequency-domain subcarrier grid.

The OFDM waveform remains the core modulation in 5G. It maps user data onto multiple orthogonal subcarriers that are independently modulated. Orthogonality ensures that subcarriers do not interfere with one another under ideal synchronization. As illustrated in Figure 2.3, the data stream is distributed across many subcarriers, each carrying a fraction of the overall sequence $X[k]$. An inverse discrete Fourier transform then converts these parallel frequency-domain symbols into a baseband time-domain signal, which is subsequently upconverted and mapped onto the time–frequency grid for transmission.

1. $T_{\text{sym}} \approx 1/\Delta f$

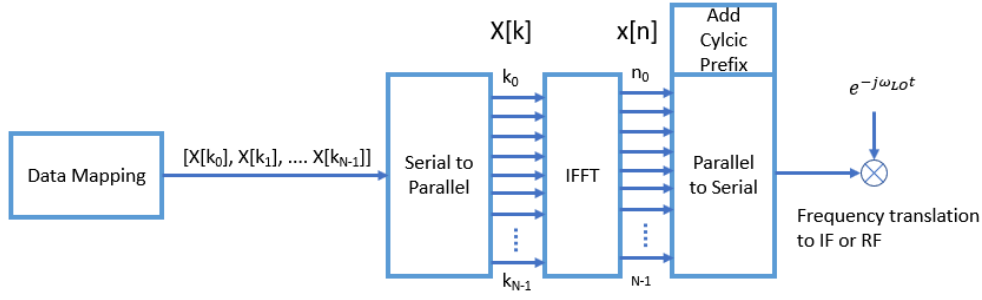


FIGURE 2.3 – Simplified pipeline of transmitter-side OFDM modulation [18].

One of the key innovations in 5G is its flexible numerology, which allows the ScS to vary with the targeted bandwidth or latency requirements. Conversely to LTE that uses a fixed 15 kHz spacing, 5G supports multiple ScS values defined by the numerology index μ , with the resulting spacing given by equation [2.1].

$$\Delta f = 15 \times 2^\mu \quad (\text{in kHz}), \text{ where } \mu \in 0, 1, 2, 3, 4. \quad (2.1)$$

As μ increases, symbols duration becomes shorter, more slots fit into a subframe, and latency can be reduced.

The resource grid is the fundamental two-dimensional structure used to organize time and frequency resources by discretizing the plane into resource elements. These resource elements are grouped into resource blocks (RBs), often referred to as physical resource blocks (PRBs), each comprising 12 consecutive subcarriers over the duration of one slot in time. This structure enables the gNodeB (gNB) scheduler to assign standardized chunks of the available time–frequency resources to different users unambiguously.

Figure [2.4] further recapitulates the relationship between numerology μ , ScS, RB bandwidth, and related timing parameters.

μ	Subcarrier Spacing $\Delta f = 2^\mu \times 15 \text{ kHz}$	Resource block bandwidth $\Delta f \times 12$ subcarriers	Frame duration	Slots per frame	Slot duration	OFDM Symbols per slot	Symbol duration	Cyclic prefix
0	15 kHz	180 kHz	10 ms	10	1 ms	14	71.43 μs	Normal
1	30 kHz	360 kHz	10 ms	20	500 μs	14	35.71 μs	Normal
2	60 kHz	720 kHz	10 ms	40	250 μs	14	17.86 μs	Normal, Extended
3	120 kHz	1.44 MHz	10 ms	80	125 μs	14	8.93 μs	Normal
4	240 kHz	2.88 MHz	10 ms	160	62.5 μs	14	4.46 μs	Normal

FIGURE 2.4 – Relationship between ScS, RB bandwidth, slot duration, etc. [42].

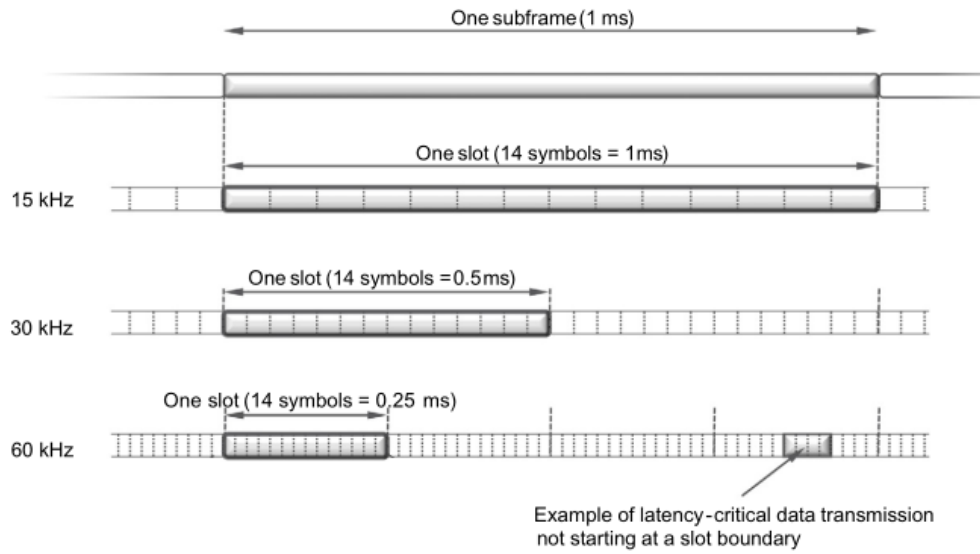


FIGURE 2.5 – Impact of the Subcarrier spacing parameter on the frame structure [24]

In summary, the design philosophy of the standard emphasizes flexibility and long-term adaptability. Instead of relying on rigid and static resource allocations, the protocol promotes the use of configurable frame structures and adjustable parameters.

2.2 Beam Management

In the context of initial access (IA), cell search refers to the process by which UE identifies and synchronizes with a suitable serving cell. For mobility purposes, cell search is performed periodically, even during an established connection, although the 5G protocol aims to minimize the set of always-on signals in order to reduce overhead and base station power consumption.

From a procedural standpoint, IA specifically concerns the transition of a UE from the idle state to the connected state. This distinction is critical, as beam management operations differ significantly depending on the UE state. In the literature, beam management often refers to operations performed when the UE is already in the connected state. In such cases, the focus is on refining the connection, monitoring channel conditions between the UE and the gNB, executing beam recovery procedures, and performing other link maintenance tasks.

In this regard, IA can be seen as a special case of beam management, one in which no prior information is shared between the two network nodes. The process typically involves probing the environment between Tx and Rx through beam sweeping over the angular sector, using synchronization signals specifically defined for initial access.

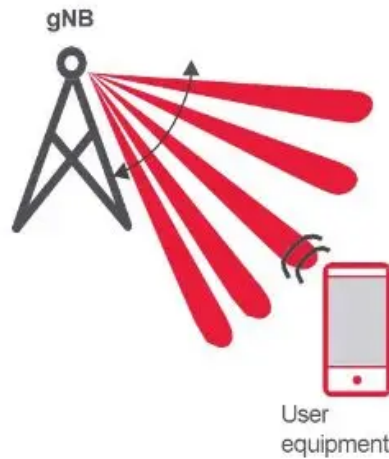


FIGURE 2.6 – gNB performing beam sweeping over a given broad angular sector [19].

According to [52], beam management encompasses the following high-level operations :

- **Beam determination** – which concerns the selection of an optimal beam from a set of candidate beams.
- **Beam measurement** – which concerns the assessment of quality metrics of the received beamformed signals, such as Layer-1 Reference Signal Received Power (RSRP).
- **Beam reporting** – which concerns the transmission of relevant measurement results from the UE to the BS.
- **Beam sweeping** – which refers to the scanning procedure that covers a given area by steering the beam across predefined angular directions.

These operations define the general functional elements of beam-based procedures but do not prescribe their exact sequencing. For example, beam reporting by the UE during IA is typically carried out via the Random Access Channel (RACH) procedure [11]. In this process, the UE initiates access to the network by transmitting a random access preamble toward the selected beam direction. This uplink transmission not only signals the UE's presence but can also carry additional control information, including beam measurement results, which the gNB uses to finalize beam alignment and establish the connection.

Alternatively, in Non-Standalone (NSA) deployments, beam reporting can also be supported via the legacy LTE anchor link. This approach allows reporting without relying solely on NR-based RACH.²

2. Nonetheless, this work prioritizes solutions that are native to 5G and do not depend on legacy systems

2.3 Synchronization Signals

2.3.1 The Synchronization Signal Block

In 5G NR, base stations periodically transmit Synchronization Signal Blocks (SSBs)³, which occupy specific time–frequency resources within the frame structure. Multiple SSBs are grouped into a Synchronization Signal Burst (SS burst), where each SSB is sequentially mapped to a predefined time position within the burst.

This transmission scheme is typically combined with analog beamforming, whereby the base station steers its beam across different angular directions and associates each SSB with a specific directional beam. In this manner, the gNB ensures full spatial coverage, allowing idle UEs to detect synchronization signals regardless of their location within the cell.

SSBs belong to the limited category of always-on signals in 5G. However, their transmission periodicity is configurable, enabling the network to reduce the constant power consumption that is required for permanent broadcast, while still offering sufficient accessibility for UEs.

In LTE, synchronization signals are always transmitted at the center of the carrier. As a result, once a device detects a synchronization signal, it automatically knows the carrier's center frequency. While straightforward, this approach has the drawback that, without prior knowledge of the carrier position in the frequency domain, the device must search across all possible carrier locations defined by the so-called carrier raster.

To accelerate the cell search procedure, 5G NR adopts a different strategy. Synchronization Signal Blocks (SSBs) are no longer restricted to the carrier center. Instead, each frequency band defines a smaller set of possible SSB positions, within the frequency domain, known as the synchronization raster. Consequently, a device only needs to perform detection over this sparser raster, rather than across the entire band. Since the carriers location and the synchronization raster responsible for the SSB position are independent, the SSB is not necessarily positioned at the carrier center and may even be misaligned with the resource-block grid.

Therefore, after detecting an SSB, the device must be explicitly informed of its exact frequency-domain location within the carrier⁴. This information is provided within the SSB itself, specifically through data carried on the third and fourth OFDM symbols of the block [26].

Frequency range	SS block frequency position SS_{REF}	GSCN	Range of GSCN
24250 – 100000 MHz	$24250.08 \text{ MHz} + N * 17.28 \text{ MHz}$, $N = 0:4383$	$22256 + N$	22256 – 26639

FIGURE 2.7 – Explicit formula of the Synchronization Raster of SSBs [5].

3. Also commonly referred to as *SS blocks*.

4. See further explanations about the Common Resource Block grid offset.

The SSB positions are known in advance by the UE, thereby reducing the search space during initial cell detection.

Each synchronization raster position is associated with a Global Synchronization Channel Number (GSCN), as shown in figure [2.7]. The GSCN identifies the frequency location of an SSB, with its reference point defined as the center frequency of the SSB. More precisely, the reference corresponds to the start of the 11th resource block of the SSB, which is equivalent to subcarrier index 121 out of the 240 subcarriers comprising the SSB. This precise definition ensures consistency in how UEs interpret and search for synchronization signals across different frequency bands.

In addition to their frequency-domain structure, SSBs are organized in time as part of SS bursts that spans a duration of 5 ms and may contain multiple SSBs [27], each mapped to different beam directions through beam sweeping. The periodicity of SS bursts is configurable, ranging from 5 ms to 160 ms. For idle mode operation, when the device has no prior knowledge, the UE assumes a default SS burst periodicity of 20 ms during the cell search procedure[26]. This assumption allows the device to determine how long it should attempt to detect SSBs at a given frequency position before switching to the next frequency in the synchronization raster.

When considering the structure of a Synchronization Signal Block (SSB), each block spans four consecutive OFDM symbols, within which three distinct signals are embedded : the *Primary Synchronization Signal* (PSS), the *Secondary Synchronization Signal* (SSS), and the *Physical Broadcast Channel* (PBCH).

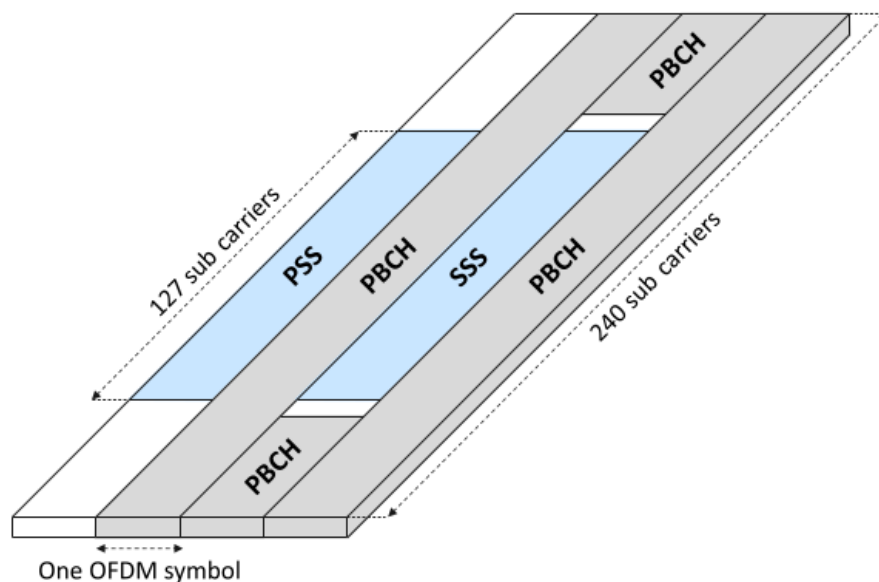


FIGURE 2.8 – OFDM symbol composition of a single Synchronization Signal Block [25].

The PSS and SSS jointly enable the UE to determine the *Physical Cell Identity* (PCI), a fundamental parameter required to uniquely identify the serving cell within a network. In parallel, the PBCH conveys the *Master Information Block* (MIB), which contains information that the UE must acquire at the earliest stage of connection. For instance,

the MIB informs the UE of the relative frequency position of the detected SSB with respect to the carrier through the *Common Resource Block* (CRB) grid offset. The CRB grid provides an absolute frequency reference that is independent of the numerology, with each CRB defined as a group of 12 subcarriers. In FR2-1, this grid is specified with a 60kHz subcarrier spacing⁵. This offset thereby establishes an unambiguous mapping of the SSB position relative to the carrier within the frequency domain.

Information	Number of Bits
SS-block time index	0 (FR1)/3 (FR2)
CellBarred flag	2
1st PDSCH DMRS position	1
SIB1 numerology	1
SIB1 Configuration	8
CRB grid offset	5 (FR1)/4 (FR2)
Half-frame bit	1
System Frame Number (SFN)	10
Cyclic Redundancy Check (CRC)	24

FIGURE 2.9 – Preliminary system information carried in the MIB [28].

In addition to the frequency positioning, the MIB provides other key parameters needed during the initial cell search and beam reporting procedures. For example, when the UE detects an SSB, it must accurately determine the corresponding SSB index in order to report this information to the base station (gNB). This association is facilitated by information elements contained within the MIB [39].

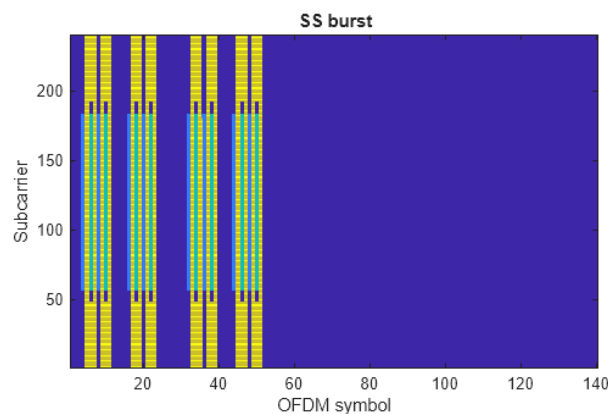


FIGURE 2.10 – Illustration of an SS burst consisting of eight sequentially transmitted SSBs [55].

5. In this work, the available transceivers operate at 28GHz, corresponding to the n257 and n261 bands of the FR2-1 range.

Each SSB spans 240 subcarriers in the frequency domain, corresponding to 20 resource blocks. Depending on the operating frequency band and the selected numerology, this allocation translates into a specific occupied bandwidth and a symbol duration, as summarized in Figure 2.11.

Numerology (kHz)	SSB Bandwidth ^a (MHz)	SSB Duration (μ s)	Frequency Range
15	3.6	≈ 285	FR1 (<3 GHz)
30	7.2	≈ 143	FR1
120	28.8	≈ 36	FR2
240	57.6	≈ 18	FR2

^aThe SS-block bandwidth is simply the number of subcarriers used for SS block (240) multiplied by the SS-block subcarrier spacing.

FIGURE 2.11 – Bandwidth and duration of SSBs as a function of numerology [26].

Because SSBs are typically transmitted with different analog beams during the sweeping procedure, they inherently extend over a temporal interval. The 5G NR specification allows up to 64 distinct SSBs within a single SS burst for the FR2-1 frequency range [50].

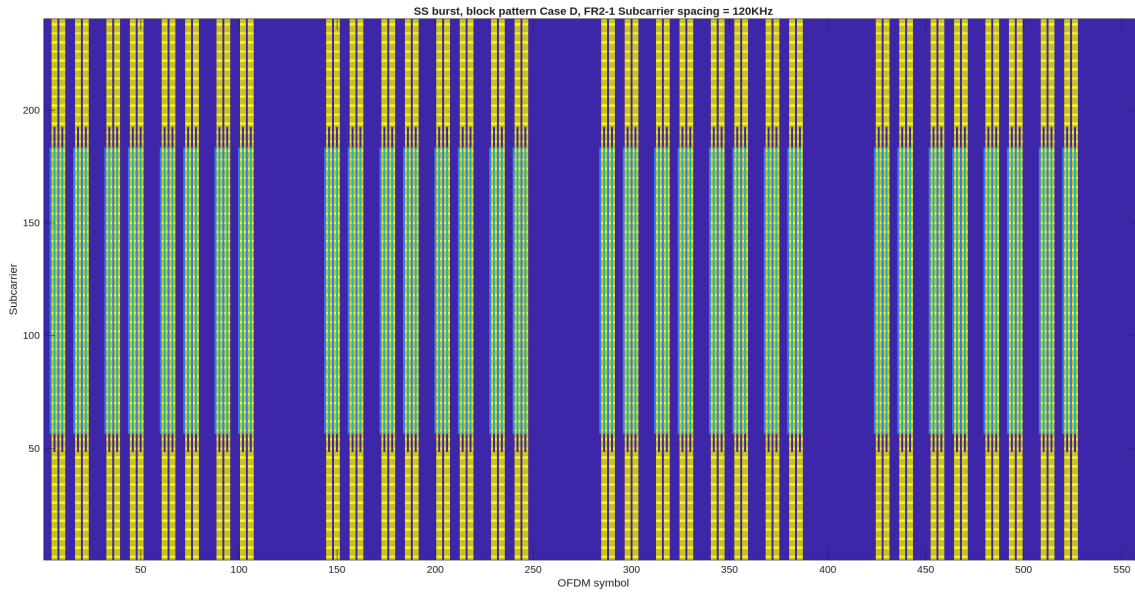


FIGURE 2.12 – Maximum case in FR2-1 : up to 64 SSBs per burst as defined in [50].

2.3.2 The Channel State Information Reference Signal

Another key signal in the NR standard is the Channel State Information Reference Signal (CSI-RS). CSI-RS plays a central role in many aspects of the standard, including channel probing, mobility measurements, interference estimation, and advanced MIMO operations. Within the context of beam establishment, CSI-RS serves as a reference signal for performing downlink measurements at the UE⁶, offering accurate evaluation of candidate beams [40].

6. The downlink refers to transmissions from the gNB to the UE, while the uplink denotes the reverse direction.

The 5G specification does not explicitly mandate whether CSI-RS or SSBs should be used for beam sweeping during initial access⁷. Nonetheless, in practice, CSI-RS is often employed in iterative beam sweeping and refinement procedures⁸. This preference arises because the number of SSBs available within a single SS burst is limited (up to 64 in FR2). Once all SSBs in a burst have been transmitted, the next opportunity to use SSB resources occurs in the next burst instant. By contrast, CSI-RS is a short signal that only spans on a single OFDM symbol and can be flexibly scheduled in both time and frequency allowing more frequent measurements.

2.4 Beam Sweeping

Up to this point, the discussion has only considered downlink signaling from the gNB to the UE. Although equivalent reference and synchronization signals also exist for uplink operation, the common assumption of beam correspondence⁹ makes it reasonable to focus on the downlink in most cases.

Within this framework, the standard also defines a set of phases that provide a reference structure for describing beam sweeping strategies [52] :

- **Phase P-1** : *"is used to enable UE measurement on different TRP¹⁰ Tx beams to support selection of TRP Tx beams/UE Rx beam(s)".*
- **Phase P-2** : *"is used to enable UE measurement on different TRP Tx beams to possibly change inter/intra-TRP Tx beam(s)".*
- **Phase P-3** : *"is used to enable UE measurement on the same TRP Tx beam to change UE Rx beam in the case UE uses beamforming".*

It is important to note that these phases are not mandatory procedural steps. Rather, they serve as a standardized reference framework that aligns naturally with hierarchical or sector-based beam sweeping approaches. The standard does not prescribe the exact resolution, granularity, or beamwidth that must be employed at each phase.

7. The standard also does not prescribe the exact procedure to implement for beam establishment

8. See section 2.4.

9. Beam correspondence is the assumption that the optimal uplink beam can be inferred directly from downlink measurements [12].

10. Transmission-Reception Point.

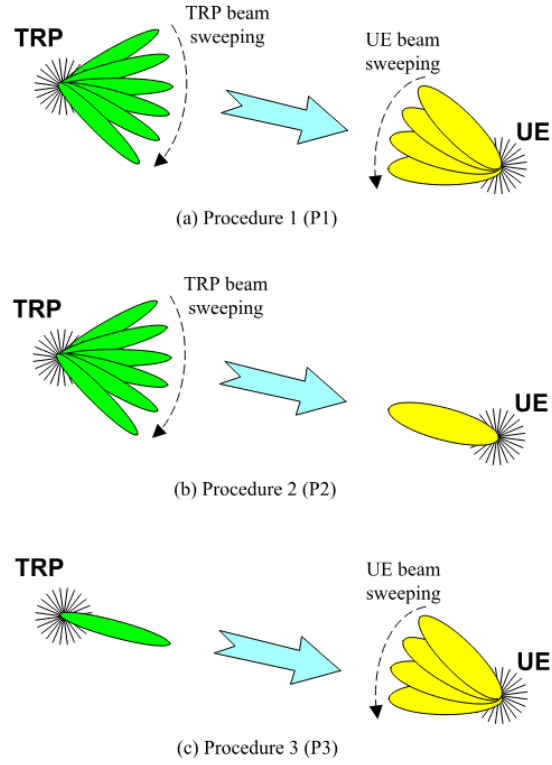


FIGURE 2.13 – Illustration of the formalized P1, P2, and P3 beam sweeping phases [40].

Having established the framework formalized by the standard, the next step is to examine the strategies and algorithms proposed in the literature for initial access and identify suitable solutions for implementation in the present work.

3 State of the Art

3.1 Beam Alignment Strategies

Numerous beam establishment solutions leverage Non-Standalone operation, where context-aided information is broadcast over sub-6 GHz microwave links to assist initial access, typically using LTE [31]. In this work, however, the focus is restricted to solutions based exclusively on Standalone mechanisms for further implementation. Within this scope, algorithms addressing the IA issue are commonly categorized into different classes [32].

3.1.1 Exhaustive Search

The most straightforward strategy for beam alignment is exhaustive search, which involves scanning the angular space across both the Angle of Departure (AoD) at the base station and the Angle of Arrival (AoA) at the user equipment. As discussed in [65], beam sweeping under the exhaustive approach is typically performed using predesigned beamforming codebooks. A codebook consists of a finite set of beam patterns, each corresponding to a specific direction obtained through predefined antenna weight configurations. By uniformly quantizing the angular domain at both transmitter and receiver, the system forms narrow beams and systematically tests all possible AoD/AoA combinations to identify the beam pair that maximizes the received power. While this method provides reliable coverage, it incurs substantial training¹ overhead and latency. Therefore, exhaustive search is often used as a baseline for comparison with more advanced schemes [65].

In [61], the authors emphasize that exhaustive search is widely analyzed under the assumption that the propagation channel remains static during the sweeping process. However, the study reminds that the exhaustive search procedure may potentially exceed the channel coherence time, thereby undermining the validity of the acquired beam alignment. Specifically, the computational load is on the order of $N_t \cdot N_r$, where N_t and N_r represent the number of BS transmit beams and UE receive beams, respectively².

Several works also explore exhaustive beam sweeping using CSI-RS. In this approach, CSI-RS resources are mapped to different BS analog beams, allowing the UE to measure and report the beam index corresponding to the strongest received signal. The BS can then align its transmission accordingly [40]. Similarly, [62] demonstrates the bene-

1. The term beam training is commonly used to denote the overall set of operations required to establish a suitable beam pair from the idle state.

2. See Table I of [61].

fits of downlink resource sweeping, noting that it avoids the need for explicit channel estimation[65].

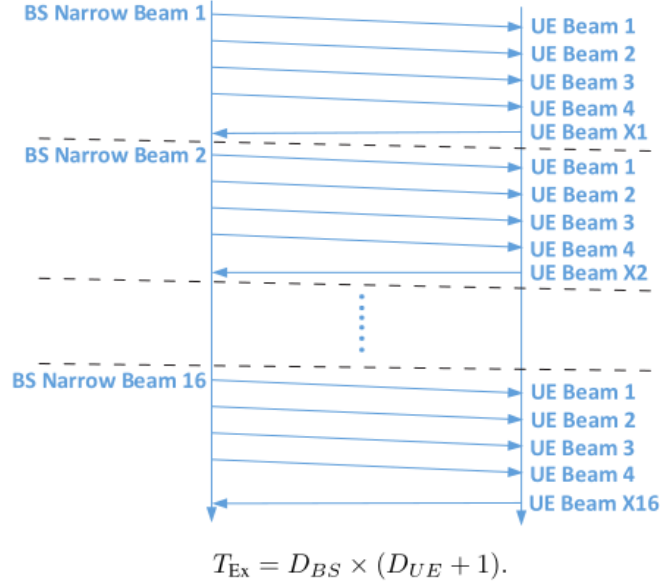


FIGURE 3.1 – Illustration of exhaustive search as described in [62], requiring $D_{UE} + 1$ beams at the UE and D_{BS} beams at the BS.

Relative to other strategies, exhaustive search often achieves superior performance in terms of misdetection probability. As illustrated in Figure 3.2, iterative and hybrid search approaches generally incur higher misdetection probabilities compared to the exhaustive method [62].

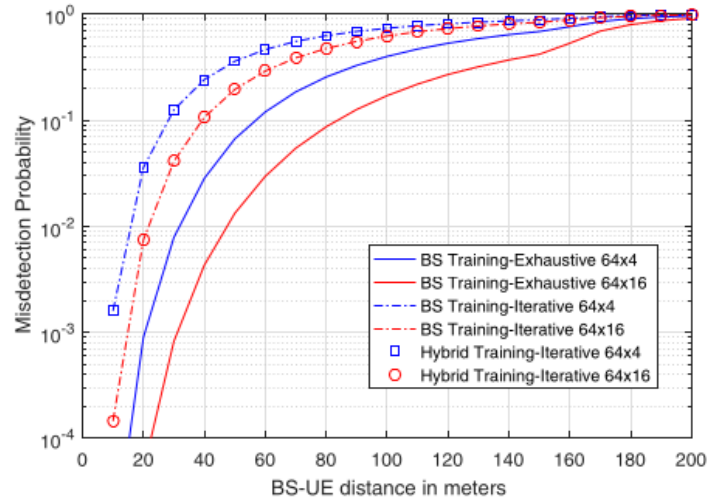


FIGURE 3.2 – Comparison of misdetection probability between exhaustive and alternative approaches [62].

3.1.2 Hierarchical Search

To mitigate the high complexity inherent to exhaustive approaches, hierarchical search is often considered as an alternative. This strategy relies on multi-layer codebooks, where beams at lower layers are wider and cover larger angular regions at the cost of reduced beamforming gain [65]. Conversely, higher-layer codebooks consist of sharper beams that provide higher gain with reduced angular coverage.

The hierarchical search process typically begins with a coarse scan of the angular domain using wide beams from the lower-layer codebook, thereby significantly reducing the number of initial transmissions. Once promising angular sectors are identified, subsequent stages refine the search using narrower beams. As a result, the performance of hierarchical search is strongly dependent on the quality of the codebook design [61][65].

An iterative hierarchical procedure based on the standardized P1–P3 phases is described in [40]. In this approach, a set of N_{Tx} transmit beams and M_{Rx} receive beams requires $N_{Tx} \cdot M_{Rx}$ transmissions during the P1 stage, where wide beams are assumed³. The subsequent P2 and P3 stages then refine the beam selection within the most promising sector identified in P1.

A similar two-stage approach is discussed in [62]. Although not fully aligned with 5G standard specifications, this method demonstrates how a hierarchical procedure can substantially reduce training overhead. Namely with a searching overhead T_{It} , that is generally smaller than that of the exhaustive search.

$$T_{It} = D_{BS1} \times (D_{UE1} + 1) + D_{BS2} \times 2, \quad (3.1)$$

$$T_{It} < T_{Ex}. \quad (3.2)$$

Despite their efficiency, hierarchical search strategies suffer from the risk of error propagation between stages. Because early stage wide beams provide relatively low gain, the dominant path may not be correctly identified. This increases the probability of misalignment in subsequent refinement stages.

3.1.3 Compressed Sensing

Compressed sensing (CS) is a signal processing technique that enables the recovery of a signal's content from a relatively small number of measurements provided that the signal is sparse.⁴ Interestingly, mmWave channels inherently exhibit such sparsity due to their limited number of significant propagation paths, which consist of the line-of-sight and a small set of multi-paths components. By exploiting this property, compressed sensing methods can be employed to efficiently identify these paths. This approach allows the estimation of good beam pair candidates with significantly reduced measurements overhead, thereby accelerating beam alignment procedures. Compressed sensing-based approaches aim to exploit the significantly reduced need for exhaustive beam sweeping, in contrast to sector scanning paradigm methods.

3. Lower-layer codebooks with wider beams are typically applied at this stage.

4. Sparsity referring here to the property of a signal having only a small subset of its coefficients carrying significant energy, while the vast majority are negligible or null.

Several article, such as [61] mention that CS is a solid option to discover the best AoD/AoA pair by identifying the non-zero elements of the environment represented in the angular space. The authors in [54] point out the limited amount of time-frequency resource allocated for initial access⁵ and propose a compressed sensing based approach compliant with the standard. In [38], the study introduces a novel beam alignment method tailored for mmWave systems in indoor environment at 60 GHz. The approach leverages the energy concentration characteristic of the discrete cosine transform in the lower-frequency coefficients to perform compressed sensing. Which enables the identification of the most dominant cluster positions, and thereby facilitates robust beamforming. The framework is model-agnostic, offers robust blockage handling, and eliminates the need for prior knowledge on the channel.

The sensing is initiated by transmitting m_1 randomly selected beams, and similarly using a random subset of m_2 receiving beams at the receiver.

$$\mathbf{Y} = \left| \mathbf{W}_{m_2}^* \mathbf{H} \mathbf{F}_{m_1} \right|^2. \quad (3.3)$$

In equation [3.3], the matrix $\mathbf{Y} \in \mathbb{C}^{m_2 \times m_1}$ contains the received signal strengths (RSS) conducted on a total of $m = m_1.m_2$ measurements beams. The matrix $\mathbf{W}_{m_2} \in \mathbb{C}^{r \times m_2}$ consists of m_2 combining vectors randomly selected from the receiver codebook, while $\mathbf{F}_{m_1} \in \mathbb{C}^{t \times m_1}$ contains the corresponding m_1 precoding vectors at the transmitter. The authors then perform a sparse representation of the rewritten measurements matrix $\mathbf{Y}$.

$$\mathbf{Y} = \left| \mathbf{S}_2 \mathbf{W}_q^* \mathbf{H} \mathbf{F}_p \mathbf{S}_1 \right|^2 = \mathbf{S}_2 \left| \mathbf{W}_p^* \mathbf{H} \mathbf{F}_q \right|^2 \mathbf{S}_1 = \mathbf{S}_2 \mathbf{\Phi} \mathbf{S}_1. \quad (3.4)$$

Here, $\mathbf{W}_q$ and $\mathbf{F}_q$ matrices consist of the complete codebooks at the receiver and transmitter, respectively. Moreover, the binary selection matrices $\mathbf{S}_1 \in \mathbb{R}^{p \times m_1}$ and $\mathbf{S}_2 \in \mathbb{R}^{m_2 \times q}$ identify the subsets of beamforming vectors randomly chosen from these codebooks. The unknown matrix $\mathbf{\Phi} \in \mathbb{C}^{q \times p}$ inherently carries the projections across all directions, and is the target of the compressed sensing recovery.

$$\text{Vec}(\mathbf{Y}) = \tilde{\mathbf{y}} = \underbrace{\mathbf{S}_2^T \otimes \mathbf{S}_1}_{\mathbf{A}} \underbrace{\text{Vec}(\mathbf{\Phi})}_{\mathbf{x}} = \mathbf{A} \mathbf{\Psi}^{-1} \hat{\mathbf{x}} \quad (3.5)$$

Equation [3.5] illustrated the vectorization step. Where $\otimes$ is the Kronecker product operator⁶, $\mathbf{\Psi}^{-1}$ is the inverse DCT transform matrix, $\hat{\mathbf{x}}$ is the vectorized $\mathbf{\Phi}$ matrix represented in the transform domain, and $\tilde{\mathbf{y}}$ is the vectorized form only containing $m < q.p$ measurements. The remaining entries are obtained by performing CS recovery and solving the l_0 -minimization problem in equation [3.6] with the Regularized Orthogonal Matching Pursuit algorithm.

$$\min \|\hat{\mathbf{x}}\|_0 \quad \text{s.t.} \quad \tilde{\mathbf{y}} = \mathbf{A} \mathbf{\Psi}^{-1} \hat{\mathbf{x}} \quad (3.6)$$

The recovered signal $\hat{\mathbf{x}}$ in the transform domain can be used to compute an approximate⁷ of $\mathbf{\Phi}$ enabling to identify clusters of higher transmitted energy in the angular

5. Referring to the maximum number of available SS Blocks per burst.

6. From the vectorization property : $\text{vec}(ABC) = (\mathbf{C}^T \otimes \mathbf{A}) \text{vec}(\mathbf{B})$ [60].

7. That is essentially a filtered version with coarser features due to high frequency DCT content removal

space.

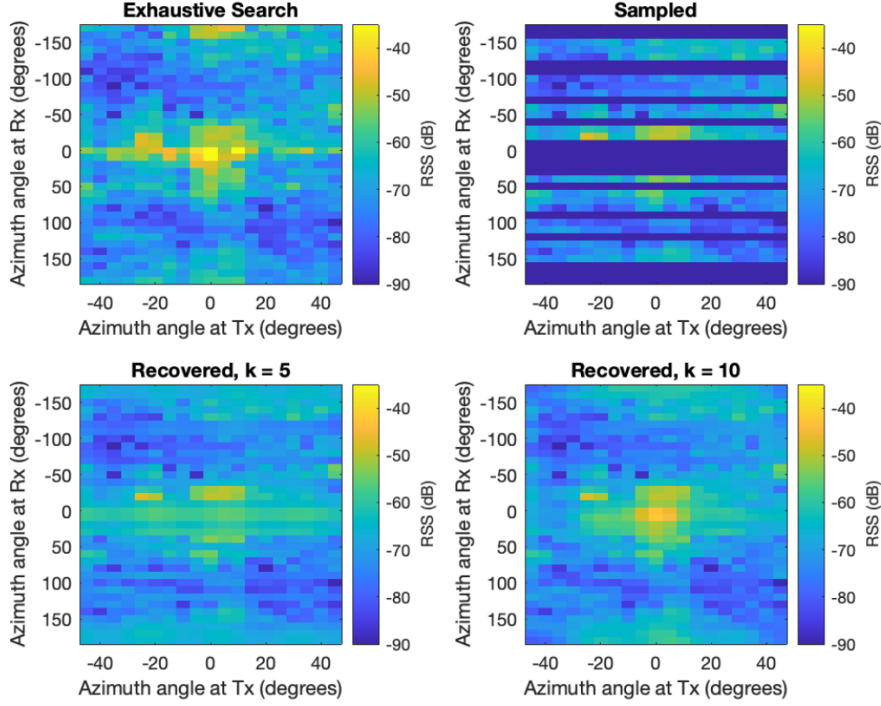


FIGURE 3.3 – Comparison between exhaustive search and CS reconstruction from sampled angular space [38]. The parameter k denotes the number of low-frequency DCT coefficients retained prior to reconstruction.

Figure 3.3 presents experimental results obtained from a 60 GHz testbed. The study demonstrates that subsampling the angular space with only 52% of the total AoD/AoA measurements, still enables accurate reconstruction of dominant angular clusters.

Another compressed sensing strategy is presented in [16], where the authors propose an algorithm that leverages compressive sensing to determine the minimum number of measurements required to identify the dominant channel cluster. The approach is combined with a gradient ascent peak detection algorithm. Specifically, the method defines a number of measurements r such that $D > r > P$, where P denotes the number of sparse channel clusters and D is the codebook length at the receiver. The study shows that the dominant cluster can be identified with computational complexity on the order of $\mathcal{O}(\log D)$.

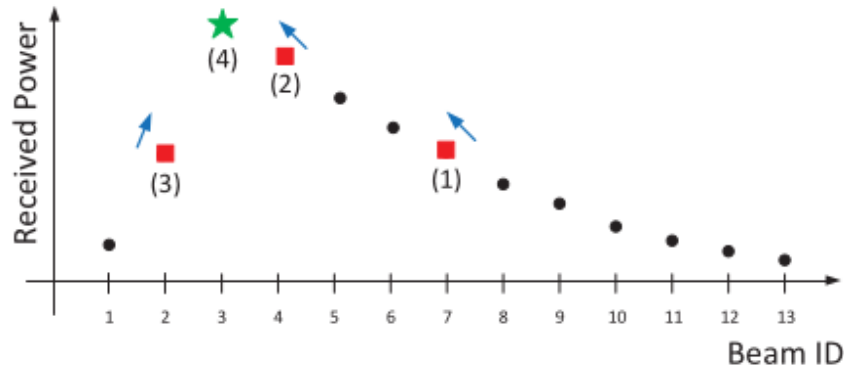


FIGURE 3.4 – Conceptual illustration of the gradient ascent procedure from [16].

Experimental validation on a custom mmWave testbed operating at 29 GHz demonstrated that the proposed algorithm reduces the beam search time by more than 90%, while maintaining a misdetection probability comparable to exhaustive search methods.

Despite these benefits, compressed sensing strategies present certain limitations. In particular, they offer limited angular resolution and reduced capability to separate closely spaced multipath components, providing instead only coarse estimates of dominant cluster regions. As a result, the reconstructed channel representation often lacks detailed features, complicating the identification of precise beam pairs. Nevertheless, within the context of initial beam establishment, such limitations may be acceptable depending on system requirements. In some cases, a slight beam pair misalignment can be tolerated provided that the resulting link budget remains within acceptable margins.

3.1.4 Artificial Intelligence

Recent growing attention has been directed toward the use of artificial intelligence (AI) techniques to improve initial access in mmWave systems. Moreover, standardization efforts have progressively incorporated machine learning (ML) concepts into the formal specifications of 5G [53]. AI-based methods exploit data-driven models to predict optimal beam pairs, thereby reducing search overhead and increasing adaptability to dynamic environments. Deep learning solutions applied to beam training have demonstrated reduced complexity, lower latency, and higher throughput [40]. Furthermore, AI models informed by parameters such as direction, velocity, and location can also enhance beam tracking by predicting UE trajectories and adapting the probing beams accordingly [34]. Finally, deep learning techniques excel at capturing non-linear relationships across angular and temporal domains [45].

The study in [21] introduces a deep neural network (DNN) model, referred to as DeepIA, designed to map a limited set of received beam measurements at the UE to an optimal transmit beam index at the base station. In practice, the UE records received signal strength (RSS) values from only a subset of the base station's transmit beams, while the model predicts the best beam among the entire set of possible transmit directions.

The DeepIA model follows a feed forward DNN architecture consisting of an input layer with M neurons, corresponding to the number of beam measurements available at the UE, and an output layer with N neurons, representing the total number of can-

didate beams at the BS. Between these, five hidden layers apply successive nonlinear transformations, as illustrated in Figure 3.5.

Layers	# neurons	Activation function
Input	$M = 2, 4, \dots, \text{or } 24$	—
Dense 1	64	ReLu
Dense 2	128	Tanh
Dense 3	256	Sigmoid
Dense 4	128	ReLu
Dense 5	64	Tanh
Output	$N = 24$	Softmax

FIGURE 3.5 – Neural network architecture and activation functions of DeepIA [21].

The dataset used for training consists of RSS samples from subsets of M beams out of the N available transmit beams. Each sample is labeled with the index of the beam most closely aligned with the receiver position, based on prior knowledge of the simulated geometry. This formulation enables supervised learning of the mapping from limited measurements to the optimal transmit beam index. Separate models are trained for different values of M , resulting in the performance curves reported in Figure 3.6.

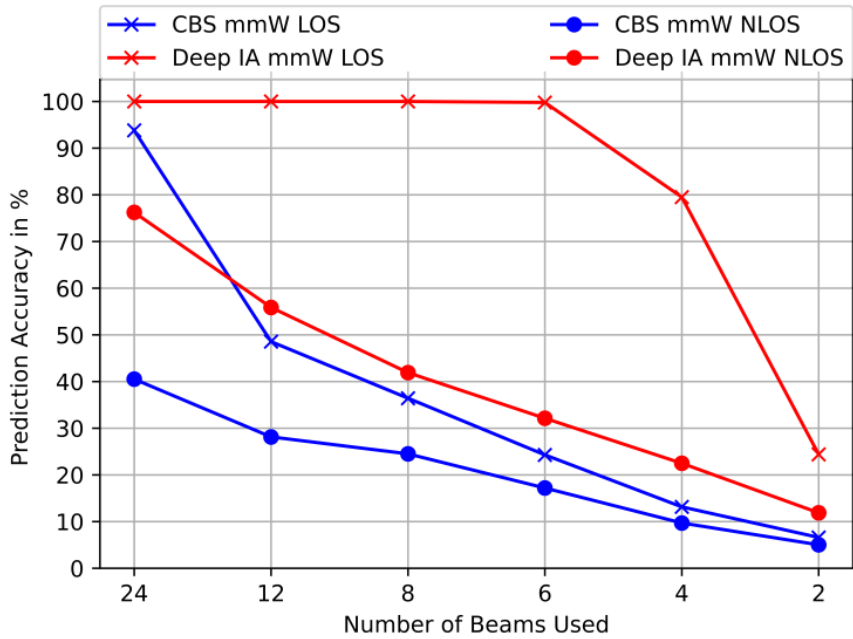


FIGURE 3.6 – Prediction accuracy of DeepIA as a function of the number of measured beams [21].

The results show that DeepIA consistently outperforms conventional beam sweeping (CBS), which simply selects the beam with the highest measured RSS. This performance gain stems from the DNN's ability to learn complex non-linear mappings, enabling it

to infer the optimal beam index even between the measured directions, unlike simple maximum selection methods that are limited to the directly observed values at the UE.

In the LoS scenario, DeepIA maintains high prediction accuracy even when the UE measures only a small subset of beams. For example, with just 4 out of 24 beams measured, the model achieves approximately 80% prediction accuracy, while CBS rapidly loses reliability with fewer measurements.

In Non-line-of-sight (NLoS) conditions, the model performance degrades more sharply, but it can still deliver substantially higher accuracy than CBS, particularly when the measurement subset remains limited.

Conversely to the approach in [21], which relies exclusively on received signal measurements, the study in [45] introduces another DNN model designed to accelerate the initial beam selection process. The method formulates beam selection as a classification problem, where the objective is to identify the most suitable beam pairs. Rather than performing scanning, the model uses the geographical coordinates of the receiver as input and produces a ranked list of the K most likely optimal beam pairs.

By restricting the initial access procedure to this reduced candidate set, the scheme significantly decreases both beam search latency and power consumption compared to conventional methods. Performance is evaluated in terms of prediction accuracy, defined as the proportion of cases in which the true optimal beam pair lies within the model's top- K predictions. The results demonstrate that the proposed DNN consistently outperforms alternative machine learning classifiers, including multi class support vector machines (SVMs) and K-nearest neighbors (KNN), which are used as benchmarks.

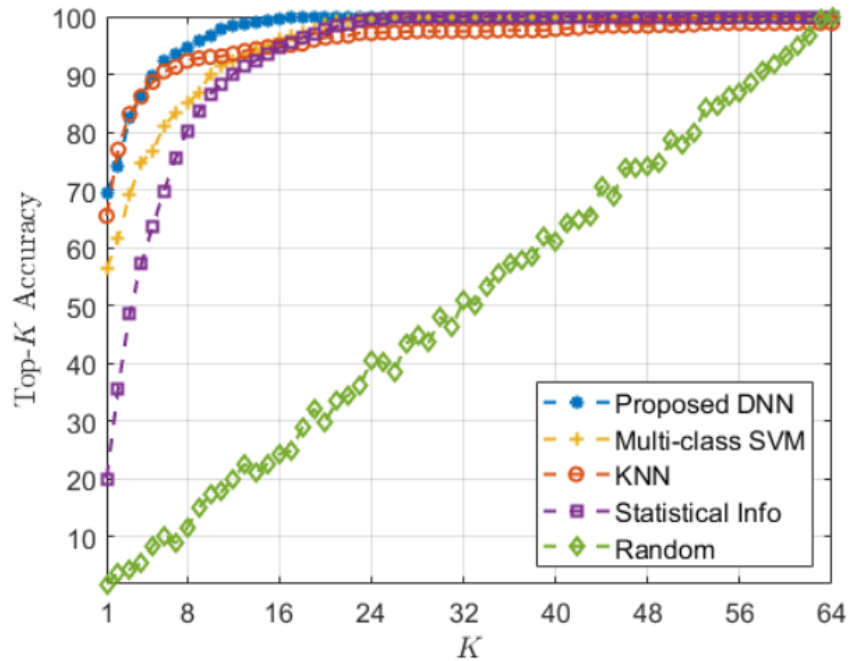


FIGURE 3.7 – Prediction performance comparison for different schemes [45].

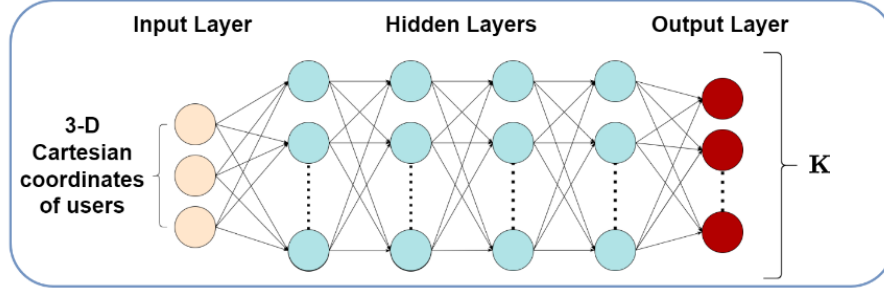


FIGURE 3.8 – DNN architecture of the proposed model in [45].

The framework was evaluated using data derived from a real mobile network topology in the n78 band⁸. Results show that an accuracy above 90% can be achieved with as few as six candidate beam pairs ($K = 6$). The accuracy remains above 90% for all scenarios with $K \geq 8$, corresponding to an eight-fold reduction in search complexity when compared to exhaustive scanning over 64 beam pairs. Specifically, for $K = 5, 8$, and 16, the reported accuracies are 89.69%, 94.73%, and 99.66%, translating into complexity reductions of 12.8-, 8-, and 4-fold, respectively. These results demonstrate that satisfactory performance can be obtained with relatively low K values, thereby reducing both search time and energy consumption, while consistently outperforming benchmark classifiers, especially at small K values as shown in figure [3.7].

Analogously, [43] proposes a data-driven beam management strategy that combines Received Signal Received Power with device orientation data extracted from an inertial measurement unit (IMU). The study introduces a deep learning model operating at 28 GHz and reports beam prediction accuracy improvements of up to 34% compared to other orientation-assisted approaches.

3.2 Selected Approaches

In the context of this work, two main strategies have been selected for practical implementation and analysis : exhaustive search and hierarchical search.

The exhaustive search method has been widely adopted in the literature and serves as a natural baseline for performance comparison. It is conceptually straightforward, and requires minimal feedback from the user equipment to the transmitter at the end of the search process. Despite its inherent high overhead, it offers reliable results and provides a consistent reference against other strategies.

The hierarchical search approach is chosen as it represents one of the most practical means to significantly reduce the search overhead while maintaining acceptable performance. This strategy does not require excessive signaling between the two ends of the link and aligns well with existing provisions in the 5G standard, which already defines procedures for iterative beam management.

Alternative strategies, such as AI based solutions, are deliberately excluded from the scope of this work. Implementing such methods would shift the focus towards data col-

8. Operating at approximately 3.5 GHz, hence not within the millimeter wave range.

lection, preprocessing, model design, and other considerations that are outside of the intended objectives. Moreover, AI assisted beam management is still relatively new in the standardization process and often relies on context-aided information, such as device position, orientation, or detailed channel state information. In contrast, the aim here is to focus on strategies that operate with minimal auxiliary information, relying primarily on reference signal measurements.

Similarly, compressed sensing approaches, while promising in certain scenarios, are not retained for this study. These methods can provide coarse environmental features and enable cluster-level estimation, but they typically struggle to deliver precise AoD and AoA pair identification. Furthermore, they impose some computational complexity on the device due to sparse recovery algorithms such as ROMP or LASSO, and their performance heavily depends on the validity of the sparsity assumptions made on the channel which not straightforward to guarantee in practical deployments.

Several evaluation criteria were considered in the literature to assess strategies. The number of beam transmissions is retained as a primary metric, as it provides a direct and easily measurable indicator to reflect the overall search effort required by each strategy. Similarly, angular deviation and Signal-to-noise ratio is also included to evaluate the precision with which the correct beam pair is identified and the quality of the received signal.

Another metric considered in this work is the Received Signal Strength. Unlike the RSRP indicator, which requires full OFDM demodulation and additional processing steps [51], RSS can be directly extracted from the reference signals with significantly lower computational effort. While RSRP is typically employed in standardized procedures, several studies reviewed above demonstrate that RSS remains an appropriate indicator for beam pair comparison. The reason for that is that the primary objective of such metrics in this context is to differentiate between candidate beam pairs and to identify the most suitable option. From this perspective, RSS provides a sufficiently reliable mean to capture the necessary information.

The access time or detection latency metric are less relevant in the present experimental context. Since the implemented setup does not perform real-time processing and feedback exchange, any timing measurement would not fully reflect practical deployment conditions. Moreover, restricting the measurement to the duration of the beam sweeping phase would yield results closely correlated to the number of beam transmissions, without providing any additional insight.

With these strategies and evaluation criteria defined, the following chapter introduces the experimental testbed, detailing the hardware components, interconnections, and practical considerations that underpin the implementation of the selected approaches.

4 Experimental Setup

For the sake of completeness, it is first necessary to outline the software environment and considerations that enable the subsequent use of the experimental hardware. All experiments are conducted under a Linux-based operating system. Before operation, a basic verification of device connectivity must be performed, ensuring that all hardware components are properly cabled and accessible by the host machine within the same local subnet. Thereafter, access to the different hardware elements relies on a set of dedicated software dependencies and libraries. Ettus Universal Software Radio Peripheral (USRP) devices require the USRP Hardware Driver (UHD) library for control and data streaming. Similarly, the TLKCore library provides the functions necessary to operate the antenna modules, while the Boost C++ library suite is used as a standard collection of utilities required during the experiments.

4.1 Hardware System Overview

Uniform planar array antennas *BBox Lite 5G* from *TMYTEK*, designed to operate on both the n257 and n261 bands, are used as basic transceivers at both ends of the transmission link. These devices convert the detected electromagnetic signal into an electrical signal, which is further conveyed to the rest of the processing chain through SMA¹ cables. The antennas are capable of analog beamforming with an azimuthal angular range of 90° centered in front of the panel. Control and configuration are managed through the *TMYTEK TLKCore* library [33]. Upon completion of the library build² [57], interoperability between the Python native tools provided by TMYTEK and C++ is achieved using Pybind11, enabling C++ access to all functionalities.

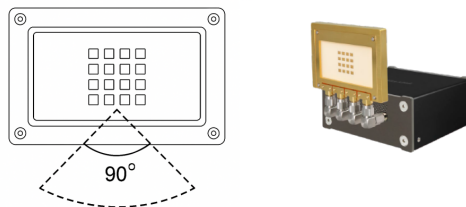


FIGURE 4.1 – Azimuth angle range capability of the BBox antenna device.

-
1. *SubMiniature version A* coaxial RF connectors.
 2. The original source files of the library are open-source and have been modified and recompiled in order to add additional functionalities.

A set of *UD Box 5G* frequency converters is used to translate the RF signal from the antennas at 28 GHz to an intermediate frequency (IF) of 4 GHz. The Tx and Rx converters, connected via SMA cables, operate with integrated mixers and are synchronized through a shared 100 MHz reference signal. Down-conversion to an intermediate frequency is a typically common practice, as subsequent stages in the processing chain is easier at lower frequencies.

The next stage of the chain is ensured by the Ettus USRP X310 software defined radio. The USRP is responsible for down-converting the IF signal from 4 GHz to baseband, sampling the signal, and streaming the resulting digital samples to the host computer. At the transmitter side, the USRP streams the baseband data provided by the host to the RF chain, while at the receiver side, it acquires the incoming samples and transmits them back to the host. This data exchange is achieved over the local network using 1G or 10G Ethernet ports, based on standard IP/UDP communication. In this sense, the USRP serves as the central interface between the analog RF chain and the digital host environment.



FIGURE 4.2 – Ettus USRP X310 device with 1G/10G ports highlighted in yellow.

For proper synchronization between the transmitter and receiver USRPs, an external clock is used to provide both a 10 MHz reference signal and a pulse-per-second (PPS) signal. The reference clock is distributed identically to both devices, from which their master clocks are derived, ensuring frequency alignment. In parallel, the PPS signal is used to synchronize the internal time registers of the USRPs, enabling accurate timing coordination during data streaming. By sharing a common reference clock, residual carrier frequency offset is minimized compared to a non-synchronized configuration.

The overall testbed is connected through a local area network, using Ethernet switches and RJ45 cables, and is managed by a single host computer. This central host ensures the control of all devices and orchestrates the experiments. In summary, the experimental setup integrates antenna arrays, frequency converters, USRP processing units, and a shared timing reference, all interconnected within a unified testbed architecture, as shown in Figure [4.3].

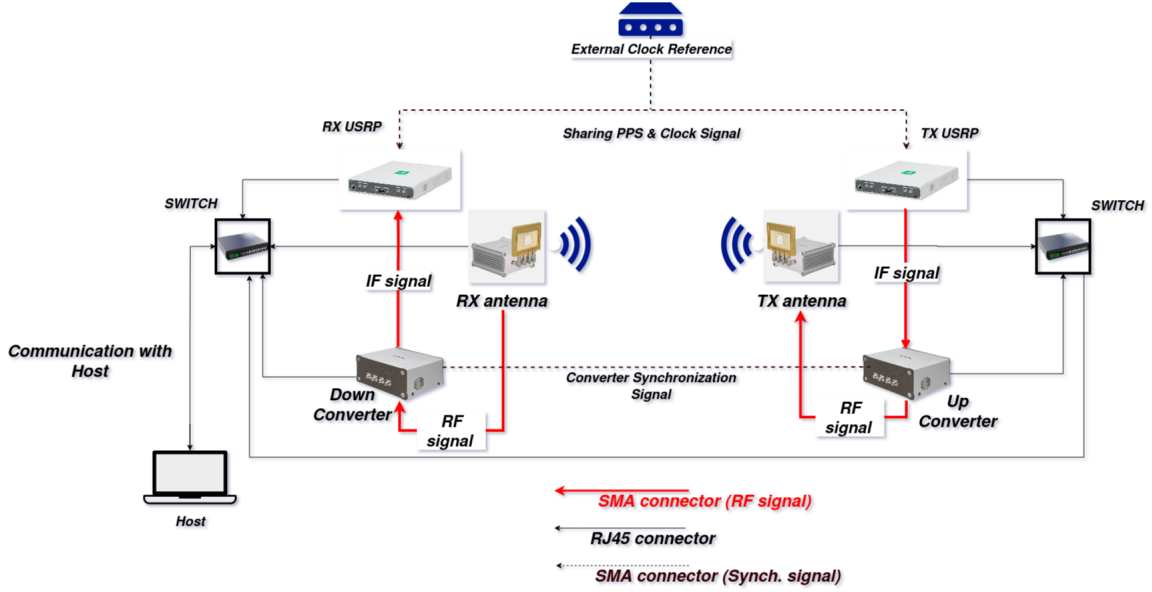


FIGURE 4.3 – Complete schematic overview of the experimental testbed.

4.1.1 Underflow and Overflow

The experimental setup is subject to *underflow* and *overflow* conditions under specific operating scenarios. During transmission, when the host is unable to sustain the data transfer rate required by the transmitting USRP, underflow occurs. In this case, the host becomes the limiting factor, as it cannot supply samples at the pace demanded by the hardware.

Conversely, overflow arises on the reception side when the host is unable to process the incoming stream of samples at the rate delivered by the receiving USRP. This results in the kernel's socket buffer being saturated, ultimately leading to packet losses. Since packet transmission and reception take place simultaneously, the effective bandwidth available for each direction is further constrained by the host's capabilities.

The USRP X310 incorporates a Xilinx Kintex-7 FPGA operated with a pre-compiled RFNoC firmware image provided by Ettus Research. This image instantiates a collection of predefined processing blocks interconnected through an internal network-on-chip (NoC)³. Interaction with these blocks, as performed in this work via the UHD RFNoC C++ API, does not involve re-synthesizing FPGA logic. Instead, it consists in configuring the parameters of the available blocks and dynamically adjusting the connection within the NoC to match the needs of the application⁴.

To address underflow at the transmitter, the *Replay Block* functionality is employed. This block allows signals to be preloaded into the FPGA's internal memory and subsequently replayed when required. By recording the waveforms of interest prior to transmission, the need for continuous real-time delivery of samples from the host is eliminated. During transmission, the block autonomously replays the samples, thereby avoiding bottlenecks. This approach, however, requires careful memory management to ensure proper

3. Available blocks include : *Digital Downconverter*, *Fast Fourier Transform*, *Radio blocks*, among others.

4. This dynamic configuration is not an FPGA re-flashing operation.

handling of the stored samples.

Despite its effectiveness in mitigating underflow, the replay block provides limited relief against overflow during reception. As illustrated in Figure 2.11, the synchronization signal blocks used for initial access occupy a relatively wide bandwidth in the FR2-1 range, resulting in high associated sample rates. While the replay block allows transmission to be carried out at maximal rates without host limitations, the reception process remains constrained by the host's processing capacity. Consequently, the sample rate selected for this study is lower than the one stipulated in the specifications. This reduction effectively compresses the usable signal bandwidth and increases symbol duration. In practice, this limitation could be alleviated by employing a more powerful desktop host, as opposed to the laptop used in the following experiments.

4.2 Software Workflow and Signal Processing

4.2.1 Software Architecture and Workflow

The experimental workflow consists of several stages. All signals of interest, as previously described, are first generated and parameterized in accordance with the 5G standard using MATLAB. These signals are then exported and preloaded into the replay memory via a C++ application running on the host computer. The executable initiates the beam search algorithm under test. During each experiment, the transmission process produces a single binary file containing all measurement data collected. Once measurements are completed, the file is processed asynchronously in MATLAB, where post-processing computations are performed to determine the best beam pair.

For hierarchical beam search algorithms, which involve multiple decision stages, feedback signaling between the transmitter and receiver is required. In such cases, the running application temporarily pauses until the MATLAB script extracts the relevant results and provides the necessary feedback to resume the experiment.

Based on the transmission parameters defining the OFDM modulation, the time-domain waveforms are generated. Figures 4.4 and 4.5 depict examples of such generated time-domain signals for CSI-RS and SSB resources, respectively⁵.

5. In the case of exhaustive search, only SSB signals are required for the experiment.

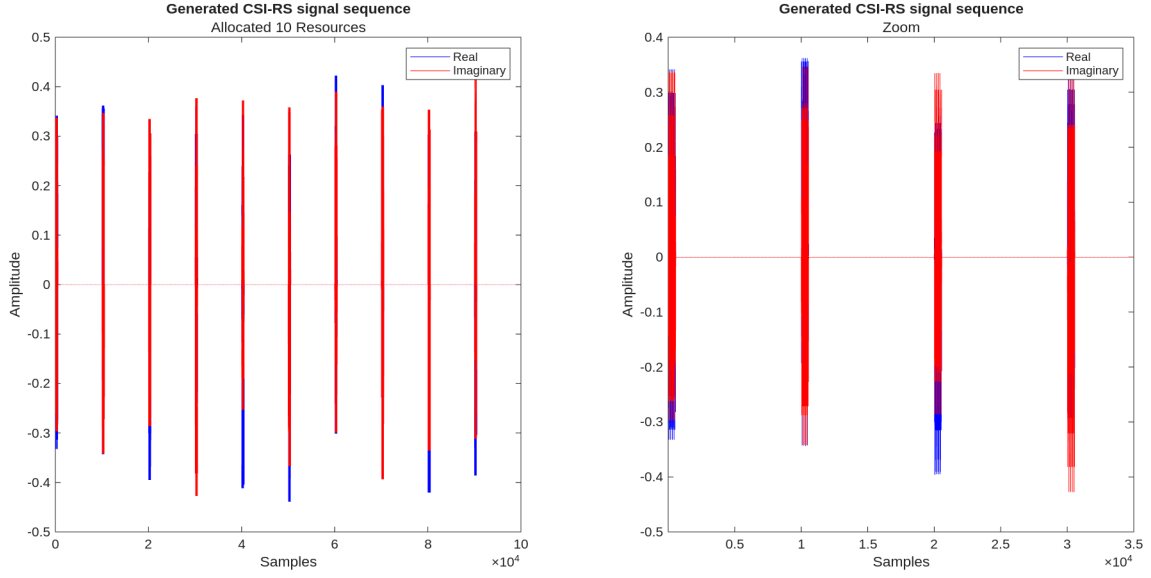


FIGURE 4.4 – Time-domain signal of generated CSI-RS resources. Subcarrier spacing is 120 kHz, with a carrier frequency f_c of 28 GHz.

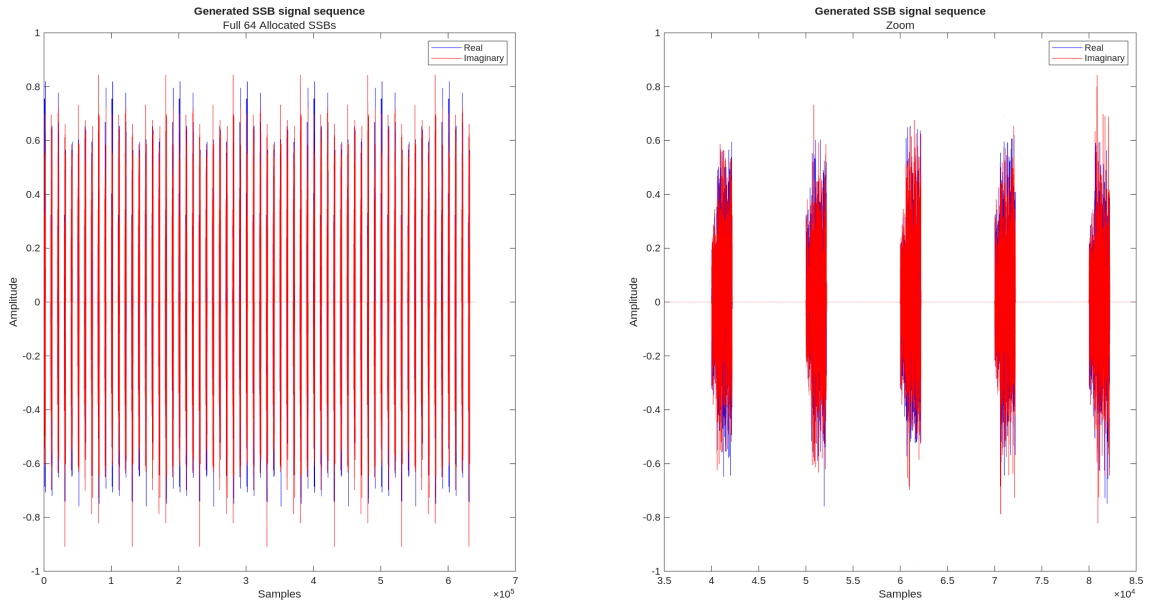


FIGURE 4.5 – Time-domain signal of generated SSB resources. Subcarrier spacing is 120 kHz, with a carrier frequency f_c of 28 GHz.

The experimental procedure is structured into two distinct phases, namely the preloading phase and the actual over-the-air (OTA) transmission. While the OTA phase depends on the specific beam alignment strategy under evaluation, the preloading phase focuses on transferring the generated signals into the USRP's internal memory for subsequent transmission.

The overall signal preloading pipeline is implemented using RFNoC utilities. The replay block is serially connected to a Digital Upconverter (DUC) block, which is responsible for frequency shifting the input signal toward the designated intermediate fre-

quency. The replay block continuously feeds the DUC with samples at a constant input rate. The DUC block, in turn, is connected to the radio block. The latter represents the processing element closest to the RF front-end and serves as the interface between the digital baseband and the IF analog domain.

Since the replay block is only used on the transmitting side, a streamer object is employed to establish the link between the host application and the instantiated RFNoC blocks on the USRP. The streamer enables the replay block to be preloaded with data directly from the input signal files prepared on the host. The replay block provides a maximum memory capacity of 1,073,741,824 bytes, which is sufficient to store the synchronization and reference signals used during the entire duration of an experiment.

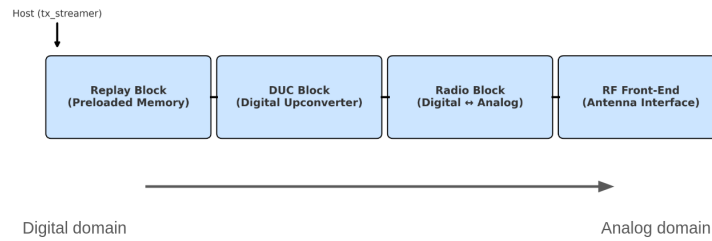


FIGURE 4.6 – Replay block preloading pipeline.

4.2.2 Acquisition and Signal Processing

After the measurements are completed, the binary output data generated during the experiments is processed using MATLAB scripts. These scripts are responsible for parsing and interpreting the raw files. The binary files contain sequences of complex baseband samples, each corresponding to the received signal for a given transmit–receive beam pair combination.

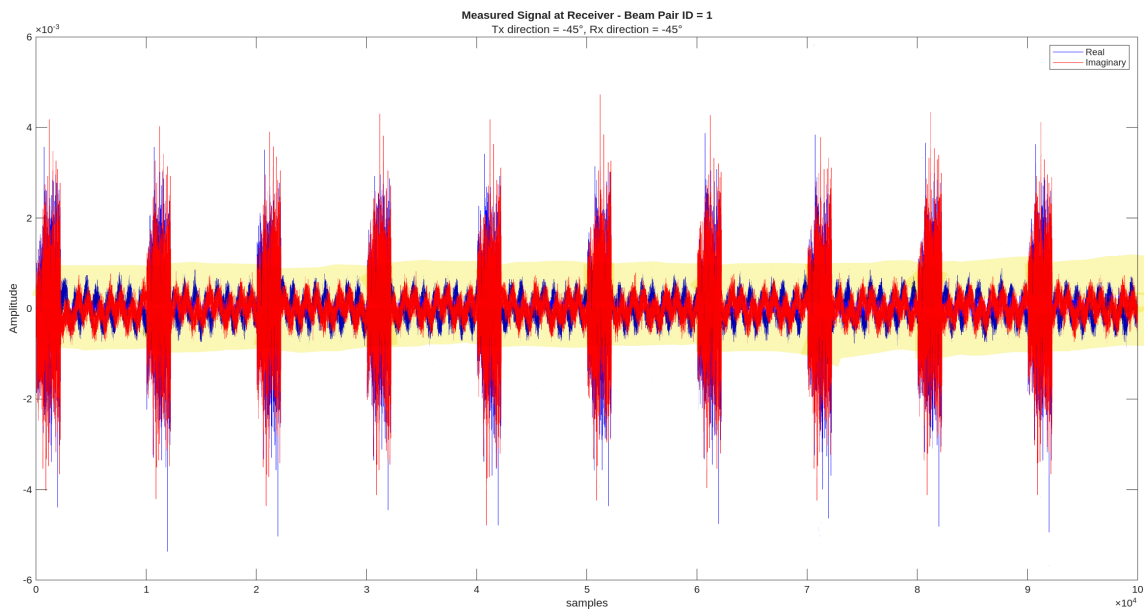


FIGURE 4.7 – Extracted signal at Rx for a given beam pair combination. Noise samples are highlighted in yellow.

Figure [4.8] demonstrates that each beam pair contains multiple replicas of the transmitted reference signal. This redundancy enables averaging across several copies, thereby reducing the impact of perturbations and improving the reliability of the measured RSS.

To accurately locate each replica within the received sequence, a cross-correlation is performed between the recorded signal and the corresponding reference signal known to have been transmitted. Since the mapping between the transmitted beam pair and the reference signal (SSB or CSI-RS) is unique, the correct pilot sequence can be selected for correlation. The resulting correlation peaks indicate the starting positions of the received replicas within the sequence.

Let $r[n] \in \mathbb{C}$, $n = 0, \dots, N - 1$, denote the recorded complex samples and let $p[k] \in \mathbb{C}$, $k = 0, \dots, L - 1$, denote a shorter known pilot sequence. For each time offset m , we correlate the pilot with the received segment starting at m and normalize by the ℓ_2 norm, as explicitly formulated in equation [4.1].

$$C[m] = \frac{\sum_{k=0}^{L-1} r[m+k] p^*[k]}{\sqrt{\sum_{k=0}^{L-1} |r[m+k]|^2}}, \quad m = 0, 1, \dots, N - L. \quad (4.1)$$

Large $|C[m]|$ magnitudes indicate strong alignment between the pilot and the received data at position m .

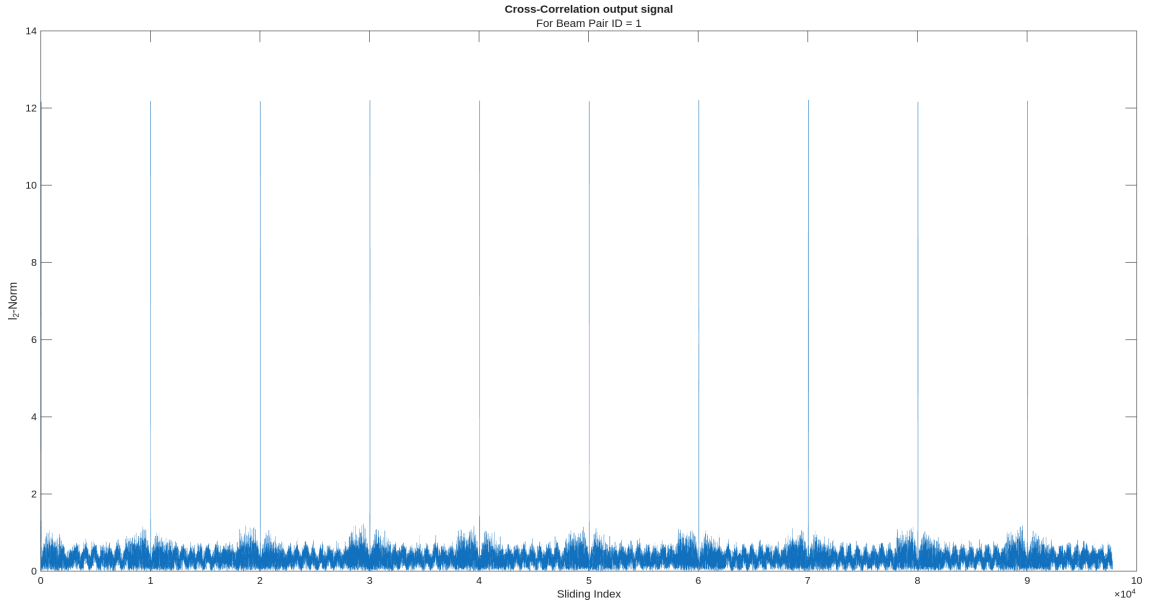


FIGURE 4.8 – Cross-Correlation output. High magnitude peaks correspond to the detected replicas of the reference signal.

After packet detection is conducted, signal of interest can be extracted from the whole sequence and the overall Received Signal Strength is derived from the fetched replica packets within the samples.

The received signal strength derived from complex baseband samples is expressed in

arbitrary digital units. In the absence of calibration, the IQ sample values are dimensionless quantities, normalized to the receiver's internal full-scale range.

Consequently, the resulting RSS values, when expressed in dB, represent relative intensity levels within this internal scale rather than absolute power. Converting these measurements into absolute power values, expressed in watts or dBm, requires a calibration procedure that maps the digital amplitudes to actual physical units, while accounting for the receiver's gain and ADC characteristics. In the present context, it is rather of less importance and these dimensionless quantities does not constitute a limitation, as the measurements share that common digital scale and are moreover used in a comparative manner regardless.

4.2.2.1 Transmission Modes and Timing Control

The default transmission software operated such that each USRP continuously executed its respective transmit and receive routines on separate threads. However, performing analog beam switching independently from the continuous transmission thread regularly led to corrupted data samples during transitions between beam pair configurations. As shown in Figure 4.9, the received signal may contain data incorrectly associated with the current beam ID, as the first replica corresponds to another beam with stronger directional alignment and higher amplitude.

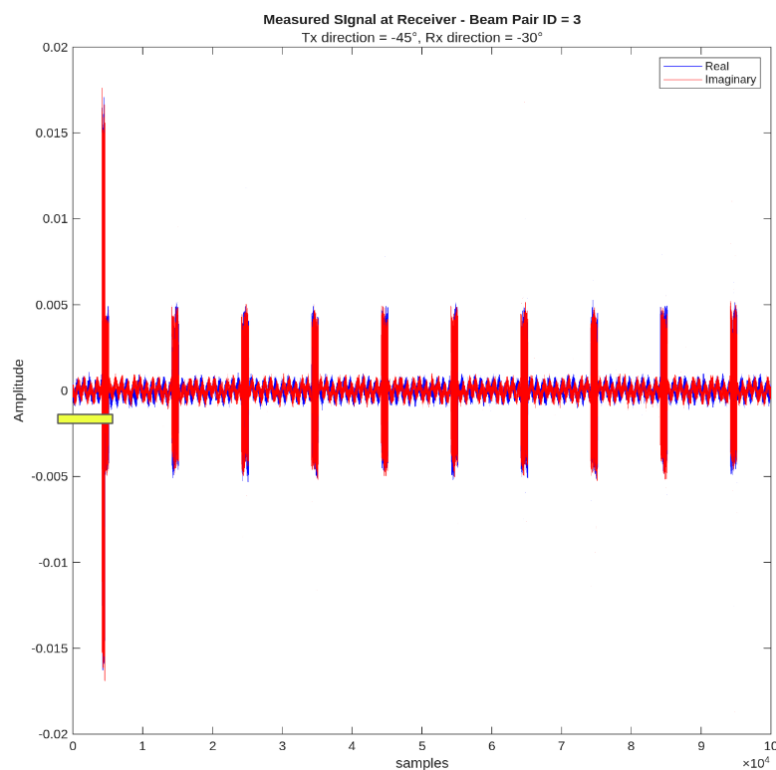


FIGURE 4.9 – Received data corruption : the first samples originate from another analog beam with higher amplitude and better alignment.

Additionally, when the replay block reloads a new reference signal, a brief transmission gap occurs. During this short interval, the receiver continues sampling, which results

in a sequence of pure noise samples, highlighted in yellow in Figure 4.9. The number of such noise samples is proportional to the latency introduced by the switching operation, which can lead to potentially several thousand samples for latencies on the order of milliseconds. Since the receiver operates continuously and independently of the transmitter, it indiscriminately records these samples, leading to corrupted datasets.

This problem is exacerbated by the heavy processing load of the receiving thread. Indeed, any unexpected software delay inflates the socket buffer and may cause overflow on the host. Overall, this independent transmission/reception mode is suboptimal, producing unreliable data samples that cannot be consistently attributed to the intended beam pair configuration.

To address these issues, transmissions were redesigned to operate in a synchronized burst mode. All instructions are managed on the main thread, and both Tx and Rx streaming commands are issued with an identical future start timestamp. This ensures that both sides begin transmission and reception simultaneously at a precise instant, and for a certain number of samples. With this method, transient effects are eliminated, as all devices are already steady and synchronized before the burst begins. The resulting received stream can thus be unambiguously associated with the corresponding analog beam configuration.

QPSK Transmission Test

To validate the effectiveness of this synchronization strategy, short bursts of QPSK-modulated packets were transmitted and analyzed. Figure 4.10 shows the received time-domain signal and the corresponding constellation.

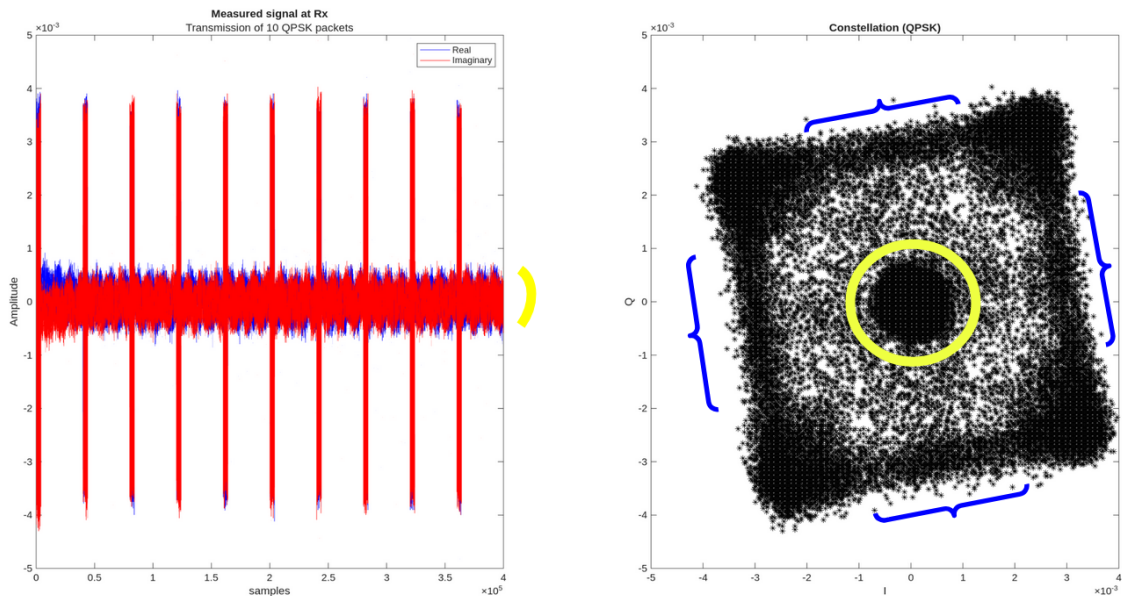


FIGURE 4.10 – Received QPSK signal. Inter-symbol sampling effects are visible in the constellation.

The time-domain signal appears correct, and the constellation exhibits clusters consistent with the transmitted QPSK symbols. Noise-only regions, highlighted in yellow, correspond to constellation points near the origin, as expected for complex Gaussian noise.

However, the constellation also reveals numerous samples located between the ideal QPSK symbol positions (indicated by the blue brackets in Figure 4.10). This misalignment is attributed to residual sampling offset between transmitter and receiver clocks, where some receiver sampling instants fall during symbol transitions.

To mitigate this, an oversampling strategy was implemented. By oversampling the received signal by an integer multiple of the transmitter's symbol rate, the temporal resolution of sampling instants is increased. This reduces the maximum timing error between the actual symbol center and the closest receiver sample. Consequently, pilot correlation peaks suffer less degradation from fractional-symbol misalignment, allowing for more reliable detection.

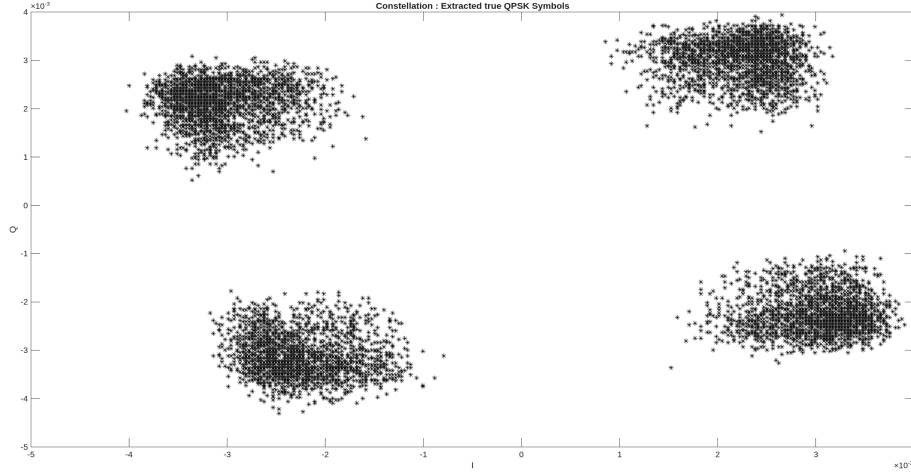


FIGURE 4.11 – Recovered QPSK constellation after oversampling. Improved temporal resolution aligns samples more closely with symbol centers.

After oversampling, the constellation recovers the expected QPSK symbol positions, with a remaining constant phase offset⁶. The conducted experiments confirm that proper synchronization and reliable reception of the transmitted signals can be achieved for subsequent measurements.

4.3 Antenna Radiation Pattern Characterization

In the context of the hierarchical approach adopted in this work, it is essential to generate radiation patterns with varying beamwidths using the available antenna modules.

The *TLKCore* library provides direct access to the RF chains of the antenna array. Each 4×4 panel comprises four independent RF chains, which can be selectively deactivated. The most straightforward method to widen the beam is therefore to intentionally deactivate part of the array, thereby broadening the radiated pattern.

To verify that the radiation patterns behave as expected under different activation configurations, measurements on the antennas were conducted. Antenna pattern characterization typically requires probing the received power at multiple incident angles. In this setup, the receiving antenna was manually rotated in steps of 5° , and the received intensity was recorded as a function of that rotation angle.

6. This residual phase offset is typically corrected during subsequent post-processing.

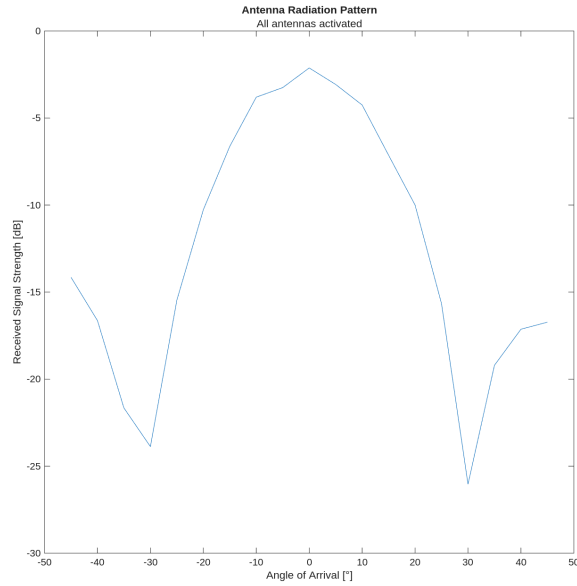


FIGURE 4.12 – Measured radiation pattern with all antenna elements activated.

Figure 4.12 shows the measured radiation pattern when all antenna elements are active, yielding the narrowest achievable beam. To better quantify the beamwidth, the same data are plotted in polar coordinates in Figure 4.13. The dashed red line indicates the -3 dB reference, from which the Half-Power Beamwidth (HPBW) can be derived.

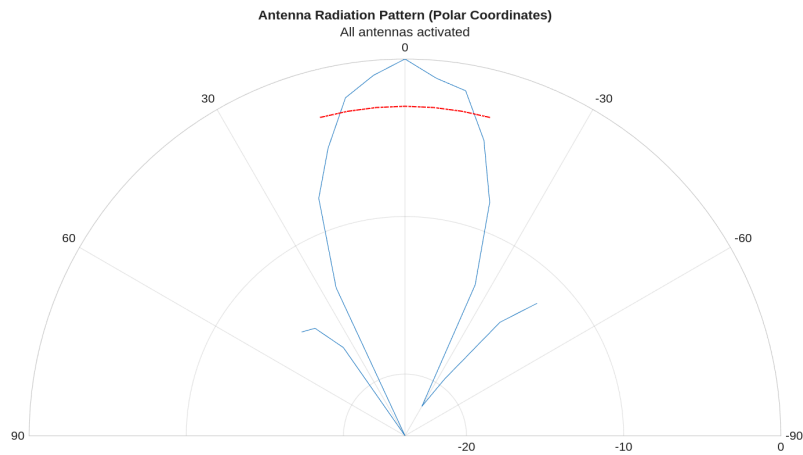


FIGURE 4.13 – Radiation pattern with all antennas activated, $\theta_{\text{HPBW}} \approx 25^\circ$.

Subsequent measurements illustrate the impact of reducing the number of active antenna elements.

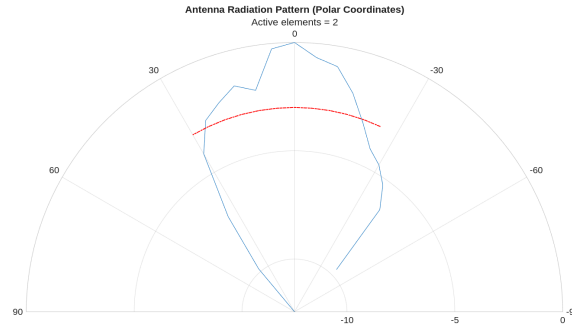


FIGURE 4.14 – Radiation pattern with 2 active elements, $\theta_{\text{HPBW}} \approx 40^\circ$.

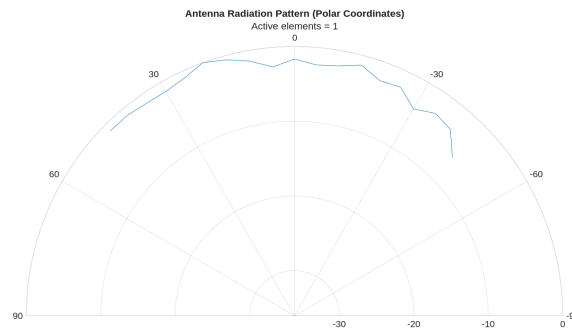


FIGURE 4.15 – Radiation pattern with 1 active element, $\theta_{\text{HPBW}} \approx 100^\circ$.

As expected, deactivating more elements results in a wider beam. In the extreme case with only one element active, the radiation pattern becomes nearly isotropic across the angular domain. Additional measurements also confirm that the beam can still be steered effectively using a reduced subset of elements, as demonstrated in Figure 4.16.

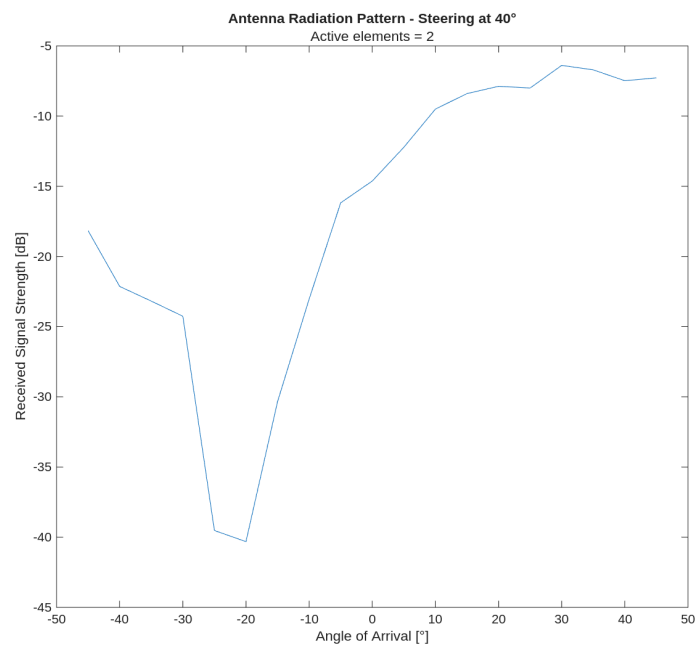


FIGURE 4.16 – Radiation pattern steered to 40° with 2 active elements.

5 Implementation and Results

5.1 Algorithms Design and Implementation

5.1.1 Exhaustive Search

The exhaustive search corresponds to a complete scan of the angular domain. In line with the 5G standard, the maximum number of SSB signals available for scanning is limited to 64, as shown in figure [2.12]. In this work, 8 beams are allocated at both the transmitter and receiver.

$$T_{Beams} = N_t \cdot N_r \quad (5.1)$$

where T_{Beams} denotes the total beam pairs, and $N_t = N_r = 8$. The choice of N_t and N_r is constrained by the antenna beamwidth θ_{HPBW} . As derived in section 4.3, the minimum achievable beamwidth is 25° , which is sufficient to cover the 90° sector with 8 beams per side. The general coverage condition is expressed in equation [5.2].

$$\theta_{HPBW} > \frac{\Delta\Phi}{N_{t,r}} \quad (5.2)$$

where $\Delta\Phi$ is the angular sector of interest.

By convention, the steerable angles are defined in $[-45^\circ, 45^\circ]$, with 0° pointing orthogonally to the array. Negative angles correspond to the right-hand side of the array, and positive to the left. The selected scan set is shown in equation [5.3].

$$\Theta_{Exhaustive} = -45^\circ, -30^\circ, -15^\circ, -5^\circ, 5^\circ, 15^\circ, 30^\circ, 45^\circ \quad (5.3)$$

All 64 beam pair combinations derived from $\Theta_{Exhaustive}$ are tested. Each pair transmits a unique SSB, allowing unambiguous identification of the best beam pair at the end of the exhaustive scan.

5.1.2 Hierarchical Search

The hierarchical strategy follows the iterative sweeping framework introduced in section 2.4 and figure [2.13]. It requires dynamic control of antenna beamwidth, achievable here with configurations of 25° , 40° , and 100° (section 4.3). To preserve link quality, the transmitter always uses the maximum number of active elements, while the receiver adapts its beamwidth per phase.

The set of steerable angles at the transmitter is represented by Θ_{Tx} and contains 6 different transmitting angular directions.

$$\Theta_{Tx} = -45^\circ, -25^\circ, -5^\circ, 5^\circ, 25^\circ, 45^\circ \quad (5.4)$$

Whereas Θ_{Rx} depends on the width configuration choice as shown in the following equation.

$$\Theta_{Rx,25^\circ} = -45^\circ, -25^\circ, -5^\circ, 5^\circ, 25^\circ, 45^\circ \quad \Theta_{Rx,40^\circ} = -40^\circ, 0^\circ, 40^\circ \quad \Theta_{Rx,100^\circ} = 0^\circ \quad (5.5)$$

Phase P1 performs a coarse scan over these sets, selecting the sector of interest. Phases P2 and P3 then refine transmitter and receiver alignment using CSI-RS signals, each phase probing 10 beams around the sector center. The final selected beam pair after P3 is retained as the hierarchical solution.

5.2 Ground Truth Scan and Performance Metrics

The measurement results presented in the following sections comprise three distinct experimental procedures. In addition to the search strategies introduced above, a third experiment is conducted. This additional approach, similar in principle to exhaustive search, performs a complete scan of the angular domain but with a substantially finer angular resolution. Specifically, the scan is carried out using steps of 3° at both the transmitter and receiver sides. Such a dense sampling results in a very large number of beam pair evaluations, making this approach impractical and non-compliant with the stringent resource constraints of 5G initial access procedures. Nevertheless, this experiment provides a comprehensive characterization of the propagation environment for a given antenna configuration and is therefore used as a *Ground Truth Scan*. Its role is to serve as a reference against which the outcomes of the other search strategies can be evaluated, allowing identification of potentially overlooked beam pair opportunities.

In addition, the previously defined performance criteria, namely angular deviation and SNR, are evaluated according to equations [5.6] to [5.10].

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \quad (5.6)$$

$$P_{\text{noise}} = \frac{1}{N_{\text{noise}}} \sum_{\text{noise region}} |e[n]|^2 \quad (5.7)$$

$$P_{\text{signal}} = \frac{1}{N_{\text{signal}}} \sum_{\text{signal region}} |r[n]|^2 \quad (5.8)$$

$$\Delta\theta = |\theta_{\text{true}} - \hat{\theta}| \quad (5.9)$$

$$\text{RSS} = \sum_{p=1}^P \left(\frac{1}{N_{\mathcal{S}_p}} \sum_{n \in \mathcal{S}_p} |r[n]| \right) \quad (5.10)$$

In the above expressions, $e[n]$ denotes the complex-valued samples containing only noise contributions, whereas $r[n]$ represents the complex-valued received samples containing both signal and noise components. The summations are performed over their respective regions within the acquired sequence of samples, corresponding to time intervals where either noise-only or signal-plus-noise is present. The division by N_{noise} or N_{signal} yields the average power for each case by normalizing the accumulated energy over the number of samples considered.

Equation [5.9] defines the angular deviation $\Delta\theta$, computed as the absolute difference between the true direction of the dominant propagation path and the estimated direction. This metric reflects the accuracy with which each method identifies the optimal beam alignment.

Finally, equation [5.10] formalizes the received signal strength (RSS) computation. Here, the index p denotes the packet number within the overall received sequence, with the outer summation iterating across all detected packets. For each packet p , the set $\mathcal{S}_p$ corresponds to the indices of the samples belonging to the packet's signal region. The inner term computes the average magnitude of the complex samples $r[n]$ within $\mathcal{S}_p$, while the outer summation aggregates these per-packet averages to obtain the overall RSS metric.

5.3 Measurements Across Scenarios

A set of distinct spatial configurations were tested, with each configuration undergoing the three beam search strategies introduced earlier. In all cases, the Tx and Rx antennas were positioned facing each other at a fixed distance of 2 meters.

The first configuration corresponds to a straightforward LoS arrangement, where the antennas are directly aligned with no obstruction. The second case also maintains an unobstructed channel, but the antennas are deliberately misaligned so that the LoS path lies near the edge of the angular domain. Finally, the third configuration blocks the LoS path entirely while introducing a reflector, creating a strong non-LoS reflection that can still be exploited as the dominant path.

5.3.1 Clear Line of Sight

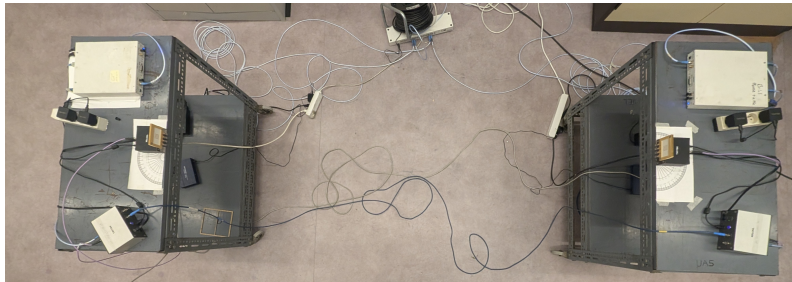


FIGURE 5.1 – First arrangement : Clear front Line-of-Sight. The transmitting antenna is on the right.

Exhaustive Search

As expected, the exhaustive search reveals a maximum received signal strength for the Tx–Rx beam pair that aligns closest to the direct LoS path. Figure [5.2] shows the time-domain baseband signal associated with the selected best beam pair, where the transmitted SSB exhibits a dominant amplitude due to the direct alignment.

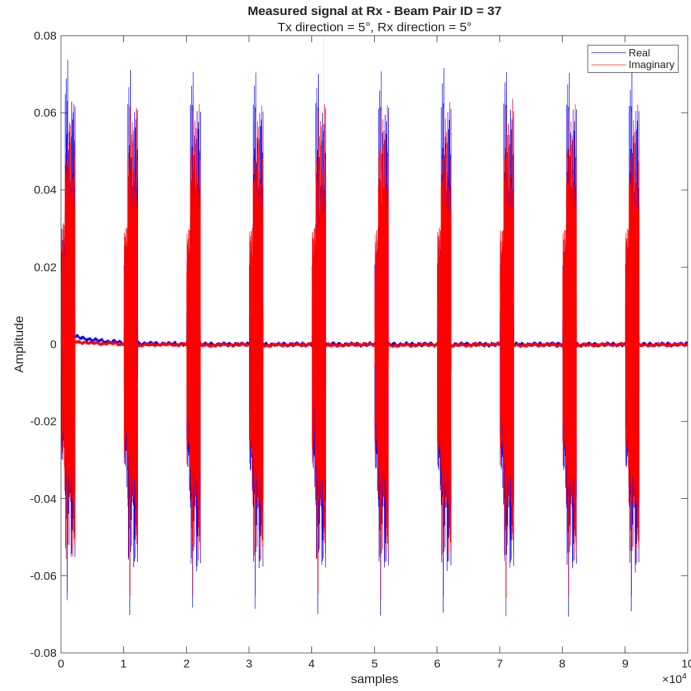


FIGURE 5.2 – Measured signal at Rx for the selected best beam pair. SNR = 33.71 dB.

Figures [5.3] and [5.4] illustrate, respectively, the measured RSS for each candidate beam pair and the corresponding 2D heatmap representation across the AoD/AoA domain. The heatmap clearly highlights a concentrated power region centered in the angular domain, which corresponds to the dominant LoS component. The spatial spread of this region is directly related to the analog beamwidth of the antenna modules.

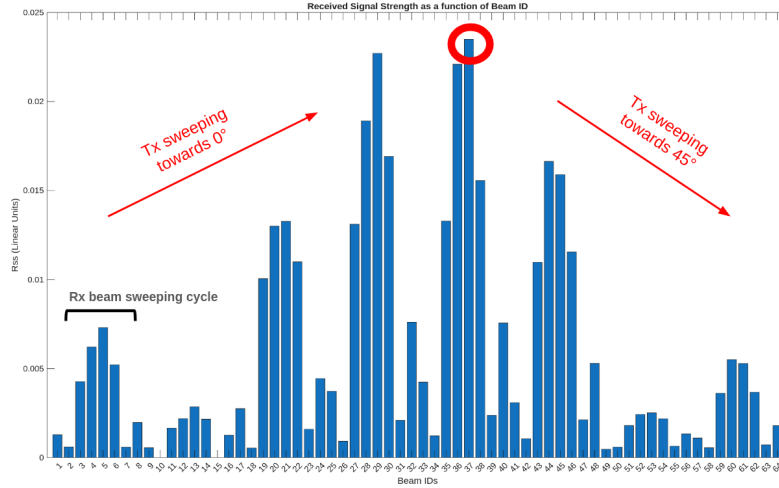


FIGURE 5.3 – Measured RSS as a function of the beam pair ID. Maximum is observed for Beam ID = 37.

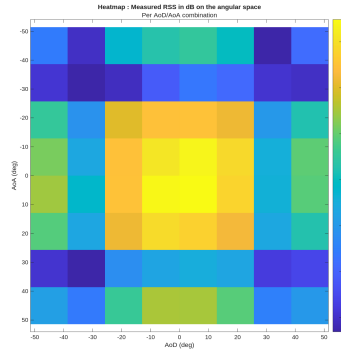


FIGURE 5.4 – RSS values measured during exhaustive search, displayed in the AoD/AoA domain (dB scale).

In Figure [5.3], the nested-loop structure of the exhaustive search is reflected in the cyclic RSS variations, with stronger regions corresponding to main lobe alignment and weaker regions corresponding to side lobes or nulls. The best beam pair was identified at $[5^\circ, 5^\circ]$ for Tx and Rx, yielding an SNR of 33.71 dB. This deviates slightly from the true geometric direction $[0^\circ, 0^\circ]$, which can be explained by the discretization of the angular grid and unavoidable small misalignments in manual antenna placement.

Hierarchical Search

The hierarchical strategy proceeds in three phases, as defined in Section 5.1.2, with results depending on the beamwidth configuration and the refinement stage.

P1

Phase 1 performs a coarse 2D angular scan, with different beamwidth configurations reducing the number of required Rx sweeps. Figure [5.5] shows the measured RSS values

for different beamwidths. Wider beams reduce scanning complexity but at the cost of lower gain, whereas narrower beams provide better angular resolution. Nevertheless, all configurations consistently identify the central angular region as the most promising sector aligned with the LoS.

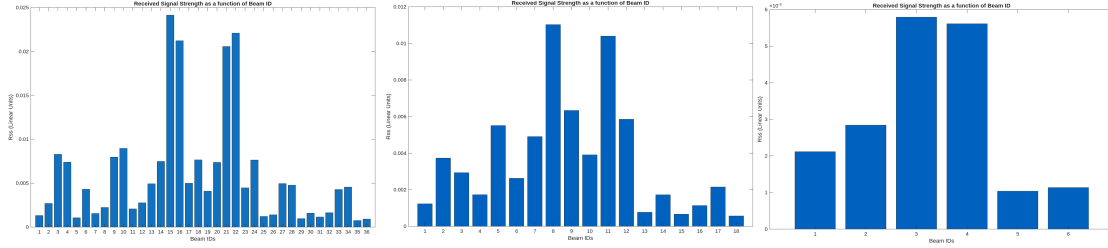


FIGURE 5.5 – RSS measurements for Phase 1 with different Rx beamwidth configurations.

Specifically :

- All antenna elements active : sector $[-5^\circ, -5^\circ]$, SNR = 32.47 dB.
- Two antenna elements active : sector $[-5^\circ, 0^\circ]$, SNR = 29.34 dB.
- One antenna element active : sector $[-5^\circ, 0^\circ]$, SNR = 25.07 dB.

These results confirm the expected SNR degradation when wider beams are used, though the relative ranking of candidate sectors remains sufficient for reliable sector selection.

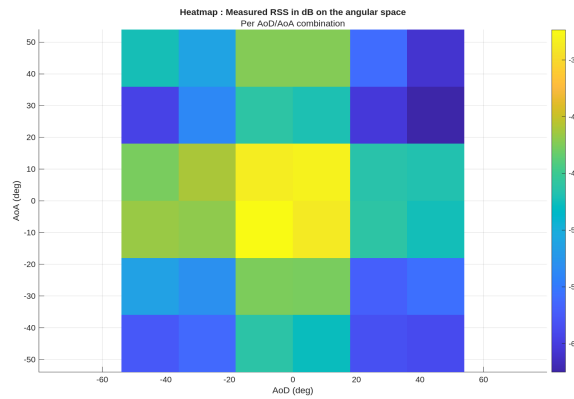


FIGURE 5.6 – Phase 1 heatmap with reduced angular resolution. The LoS path remains identifiable.

P2

Phase 2 refines the Tx beam within the selected sector, now using CSI-RS resources. Figure [5.7] shows the resulting RSS measurements. Due to the limited minimum beam-width (approximately 25°), adjacent beams strongly overlap, making RSS variations relatively small and more difficult to discriminate. Despite this limitation, small differences still enable selection of the most suitable Tx direction.

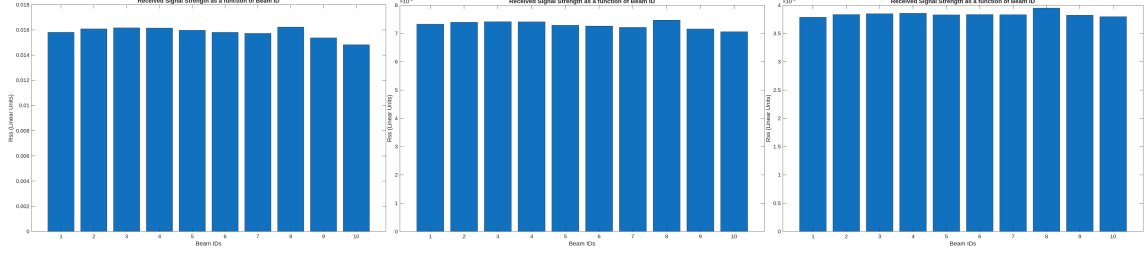


FIGURE 5.7 – Phase 2 RSS values for different beamwidth configurations. Overlap between adjacent beams reduces RSS variability.

P3

Phase 3 further refines the Rx beam within the selected sector, still based on downlink transmissions. Results for different configurations are summarized below :

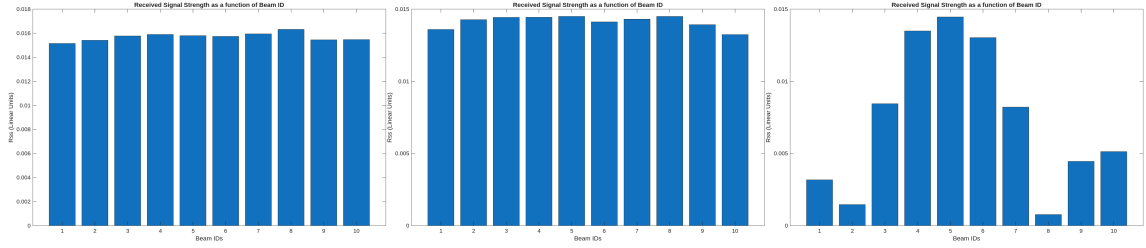


FIGURE 5.8 – Phase 3 RSS values during Rx refinement.

- All antenna elements active : beam pair $[-2^\circ, -2^\circ]$, SNR = 28.84 dB.
- Two antenna elements active : beam pair $[-2^\circ, 0^\circ]$, SNR = 28.39 dB.
- One antenna element active : beam pair $[-2^\circ, 0^\circ]$, SNR = 28.40 dB.

All configurations converge to nearly identical beam pairs close to the LoS path, with angular deviations not exceeding 2° . Since refinement stages are performed with all antenna elements active, the final SNRs are comparable, regardless of the initial P1 configuration. This demonstrates the efficiency of wider beams in the early stages, as they provide significant complexity reduction while still converging to near-optimal results.

Ground Truth Scan

The ground truth scan serves as a reference to assess the validity of the other approaches.

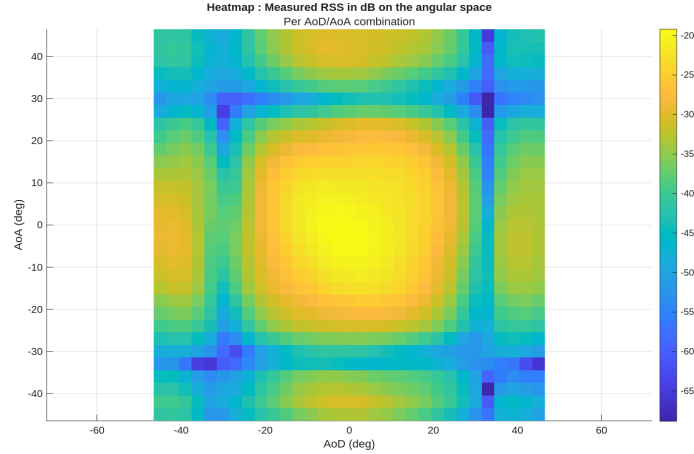


FIGURE 5.9 – Ground truth scan heatmap. Secondary peaks originate from side-lobe radiation.

As shown in Figure [5.9], the LoS component clearly dominates. Secondary local maxima appear at the angular domain edges, which are attributed to side-lobe energy rather than genuine multipath components. The ground truth scan identifies the best beam pair at $[-3^\circ, 0^\circ]$, again slightly deviating from the expected $[0^\circ, 0^\circ]$ due to discretization and small physical misalignments. This underlines that, in practice, increasing the angular sampling density does not necessarily yield perfectly aligned results, as residual errors inevitably stem from manual placement inaccuracies and analog beam steering limitations.

5.3.2 Near-Edge Line of Sight

This spatial configuration purposely misaligns the antennas so that the dominant LoS path can only be observed near the boundary of the angular domain. The goal is to evaluate whether the proposed strategies remain robust when the direct path is located at the extreme edges of the scanning range, where antenna gain is typically reduced and detection becomes more challenging.

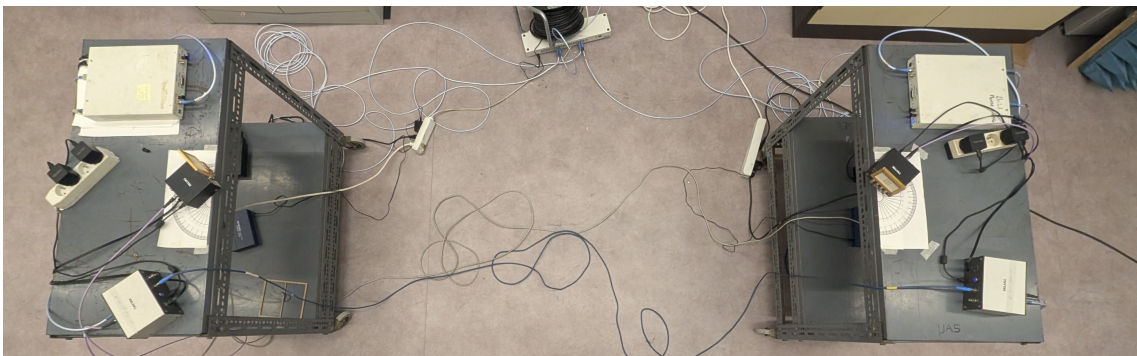


FIGURE 5.10 – Second arrangement : LoS component located at the angular domain boundary.

Exhaustive Search

Figure [5.11] shows the exhaustive search heatmap, where the region of maximum power is now concentrated near the domain boundary, at approximately $[-45^\circ, -45^\circ]$. This confirms the ability of exhaustive sweeping to identify the dominant path even under reduced antenna gain conditions at the angular limits.

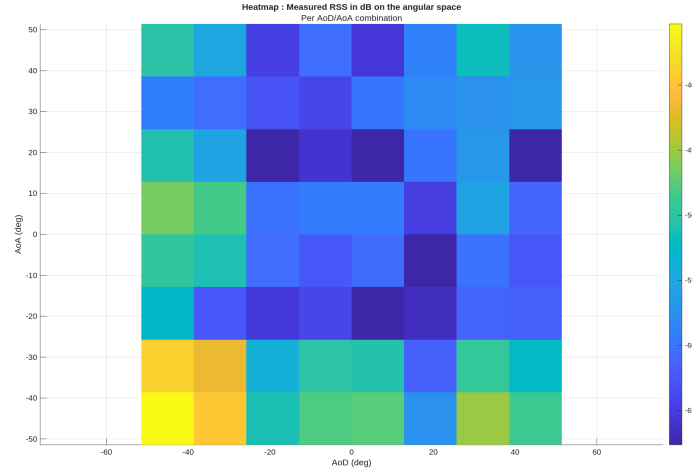


FIGURE 5.11 – Heatmap of RSS values obtained during exhaustive search in the near-edge LoS configuration.

The corresponding RSS evolution over the scanning procedure is depicted in Figure [5.12], where the dominant peak clearly emerges at the edge of the tested beam pairs.

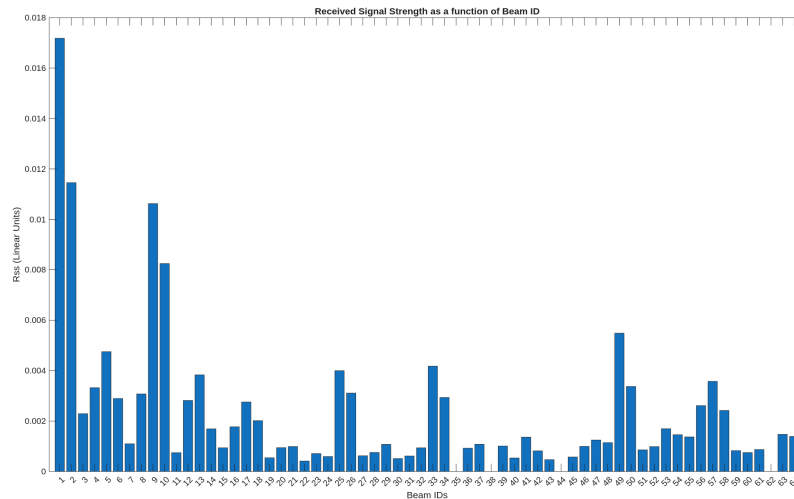


FIGURE 5.12 – Measured RSS during exhaustive search for the near-edge LoS configuration.

The final selected beam pair corresponds to $T_x = -45^\circ$, $R_x = -45^\circ$, yielding an SNR of 31.74 dB.

Hierarchical Search

As with the previous arrangement, the hierarchical search may suffer from the reduced angular resolution in its early stages, particularly under wider Rx beam configurations. Nevertheless, the edge placement of the LoS path enables the strategies to converge with relatively good accuracy.

P1

The results of Phase 1 are summarized as follows :

- All antenna elements active : $[-45^\circ, -45^\circ]$, SNR = 31.83 dB.
- Two antenna elements active : $[-45^\circ, -40^\circ]$, SNR = 27.90 dB.
- One antenna element active : $[-45^\circ, 0^\circ]$, SNR = 21.03 dB.

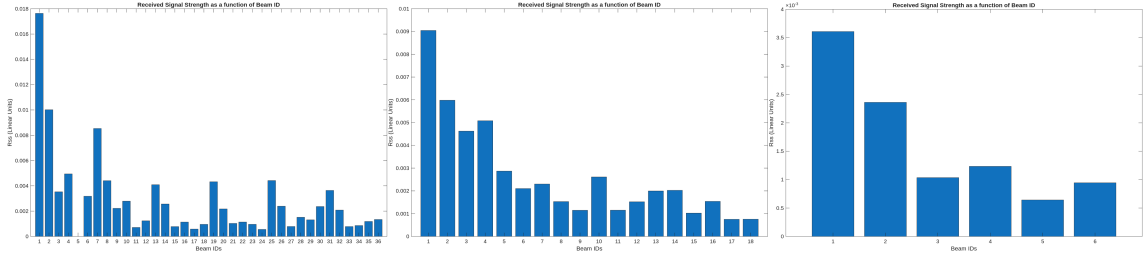


FIGURE 5.13 – Phase 1 RSS measurements for different Rx beamwidth configurations in the near-edge LoS case.

The results indicate that Phase 1 is not heavily penalized by coarser Rx beamwidths in this configuration, since the LoS component remains prominent at the boundary. A clear trend is observed, early beam candidates, corresponding to angles near the edge of the domain, exhibit higher RSS values allowing for reliable sector selection. As the surrounding environment remains mostly at the noise level, with no detectable multipath components, the LoS path stands out clearly. This distinct separation enables P1 to reliably identify the appropriate angular sector for subsequent refinement.

P2

The RSS measurements associated with P2 are presented in Figure [5.14]. Although adjacent beam overlap reduces the discriminative power of the RSS metric, the system remains able to identify candidate Tx beams aligned with the dominant path.

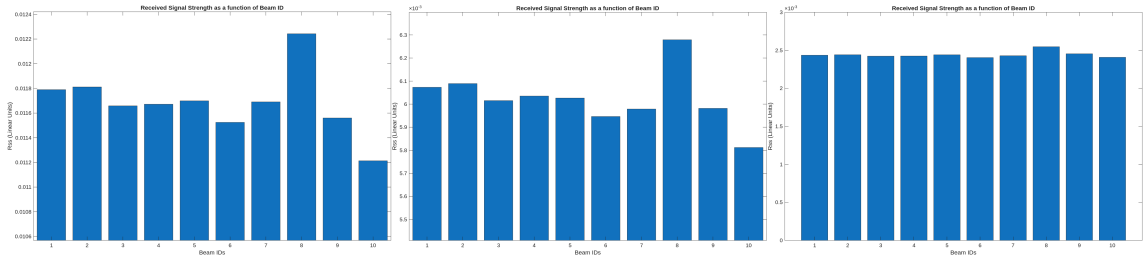


FIGURE 5.14 – Phase 2 RSS measurements for different beamwidth configurations in the near-edge LoS scenario.

P3

Finally, Phase 3 performs refinement at the Rx side within the selected sector. The results are summarized as follows.

- All antenna elements active : $[-42^\circ, -42^\circ]$, SNR = 27.91 dB.
- Two antenna elements active : $[-42^\circ, -42^\circ]$, SNR = 27.62 dB.
- One antenna element active : $[-42^\circ, -40^\circ]$, SNR = 27.99 dB.

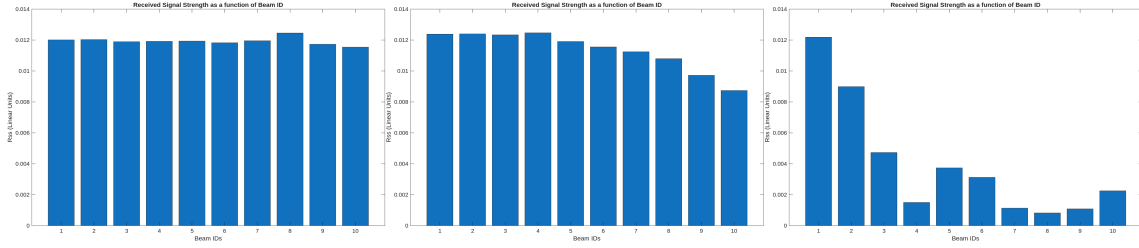


FIGURE 5.15 – Phase 3 RSS measurements for different Rx beamwidth configurations in the near-edge LoS scenario.

Similar to the central LoS case, all configurations converge towards beam pairs close to the true dominant path. However, the initial beamwidth choice at the Rx has a clear influence : configurations starting with wider Rx beams simplify sector detection in Phase 1 but reduce the precision of the final Rx beam refinement, often leading to slightly larger angular deviations. Conversely, narrower beams offer finer precision but require more resources.

Ground Truth Scan

The ground truth scan confirms the presence of a dominant path near the boundary of the angular space, as shown in Figure [5.16]. Secondary maxima are observed, largely attributable to side-lobe radiation.

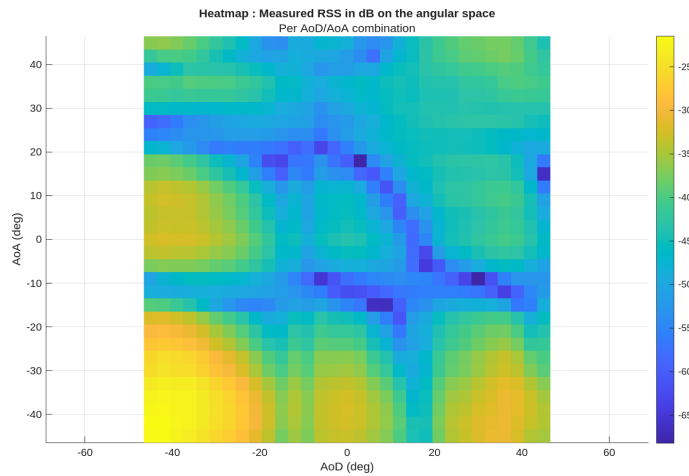


FIGURE 5.16 – Ground truth scan heatmap for the near-edge LoS arrangement.

The ground truth scan ultimately identifies the optimal beam pair as $T_x = -42^\circ$, $R_x = -42^\circ$, consistent with the hierarchical results and confirming the reliability of the strategies in boundary conditions.

5.3.3 Non Line-of-Sight with Reflector

In contrast to the previous LoS-based configurations, this experiment introduces a physical layout in which the direct path is intentionally obstructed. Communication between Tx and Rx is therefore only achievable through a single controlled reflection path. A metallic reflector was precisely positioned to ensure that the transmitted wave could be redirected towards the receiver under the expected geometry.

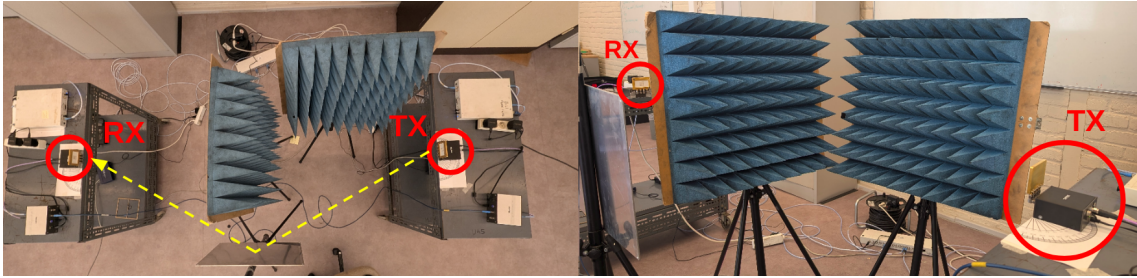


FIGURE 5.17 – Third arrangement : LoS obstructed, with communication established via a metallic reflector.

The reflector position was derived using basic trigonometric relations, as illustrated in Figure [5.18]. For Tx–Rx separation d , and a desired incidence angle α , the reflector offset L must satisfy $\tan(\alpha) = \frac{L}{\left(\frac{d}{2}\right)}$.

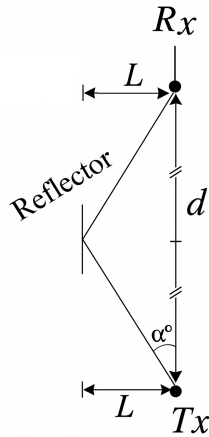


FIGURE 5.18 – Simplified geometry for reflector placement.

With $d = 2$ m and the arbitrarily fixed $\alpha = 30^\circ$, the resulting offset is $L = \tan(30^\circ) \approx 0.58$ m. Accordingly, the reflector was positioned at 58 cm from the midpoint of the link.

Anechoic absorber panels were added to obstruct any alternative LoS or multipath components, ensuring that the reflected path was the sole viable option.

Exhaustive Search

The exhaustive search correctly identified the reflected path as the dominant communication link.

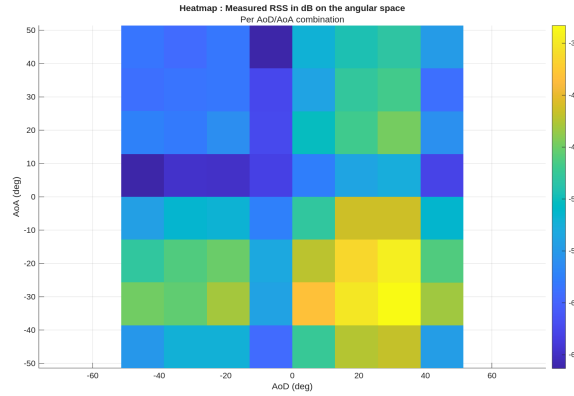


FIGURE 5.19 – Exhaustive search heatmap for the reflector-based NLoS scenario.

The selected beam pair corresponds to $T_x = 30^\circ$, $R_x = -30^\circ$, yielding an SNR of 31.92 dB. At first glance, this appears to be a perfect match with zero angular deviation. However, this outcome is partly explained by the discretization of the tested angular grid : since 30° was itself one of the available candidate directions, the exhaustive sweep unsurprisingly converged exactly on it.

Hierarchical Search

P1

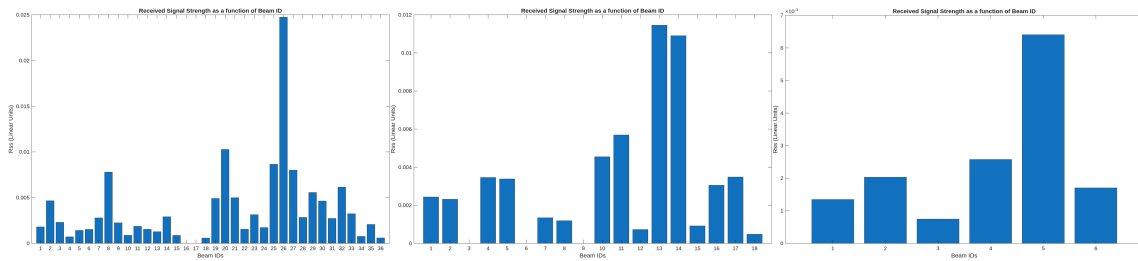


FIGURE 5.20 – Phase 1 RSS measurements for different Rx beamwidth configurations in the NLoS scenario.

Phase 1 sector selection yielded the following results :

- All antenna elements active : $T_x = 25^\circ$, $R_x = -25^\circ$, SNR = 32.17 dB.
- Two antenna elements active : $T_x = 25^\circ$, $R_x = -40^\circ$, SNR = 30.13 dB.
- One antenna element active : $T_x = 25^\circ$, $R_x = 0^\circ$, SNR = 25.87 dB.

P2

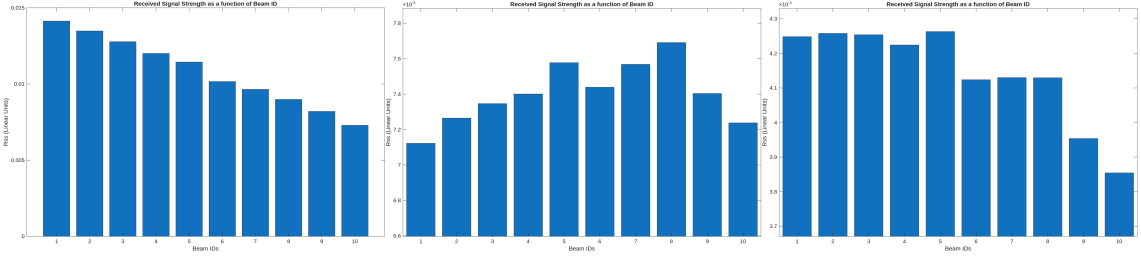


FIGURE 5.21 – Phase 2 RSS measurements for different beamwidth configurations in the NLoS scenario.

Unlike in the LoS scenarios, Phase 2 exhibits clearer RSS discrimination among candidates. This is explained by the geometry of the reflection path, which strongly constrains the set of viable angles. Beams not aligned with the reflector contribute very little power, making the correct candidate easier to identify.

P3

The Phase 3 refinement converged towards beam pairs closely aligned with the reflector-induced path :

- All antenna elements active : $T_x = 31^\circ$, $R_x = -29^\circ$, SNR = 29.04 dB.
- Two antenna elements active : $T_x = 28^\circ$, $R_x = -32^\circ$, SNR = 29.46 dB.
- One antenna element active : $T_x = 25^\circ$, $R_x = -30^\circ$, SNR = 29.58 dB.

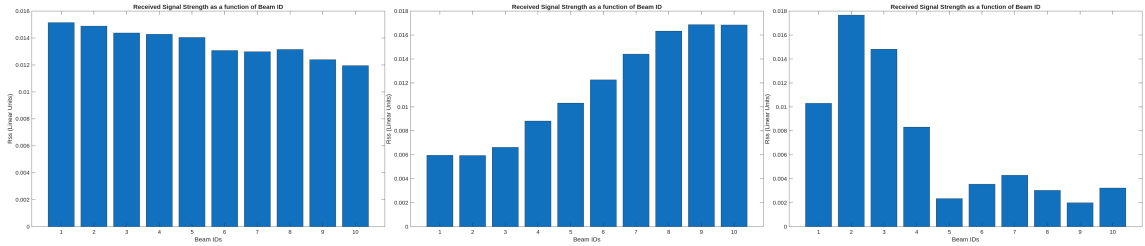


FIGURE 5.22 – Phase 3 RSS measurements for different beamwidth configurations in the NLoS scenario.

The three configurations converge to beam pairs centered around the reflection angle, though with small deviations depending on Rx beamwidth. As in the previous experiments, wider beams at Rx introduce larger angular errors at the reception beam. Nevertheless, the resulting SNR values remain comparable across configurations, indicating that reliable link establishment is possible even with imperfect angular precision and wider beam setup. Regarding the overall achieved SNR, even with only a single element activated during the early stages of the hierarchical approach, the configuration is still able to deliver satisfactory results, and provide significant overhead reduction. This is consistent with the relatively broad beam shapes of the hardware, which allow substantial

power capture even when the beams are not perfectly aligned with the reflector. With finer beamforming resolution (e.g., pencil beams), such deviations would likely have a stronger impact of the received signal quality and the SNR.

Ground Truth Scan

The ground truth scan, presented in Figure [5.23], confirms the existence of a strong reflected path near the expected geometry.

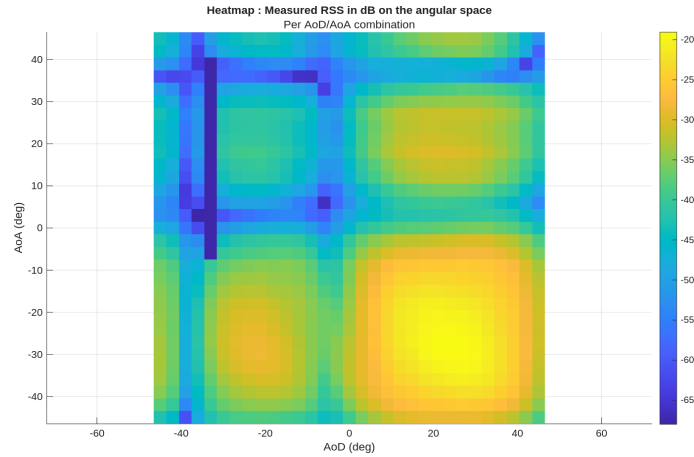


FIGURE 5.23 – Ground truth scan heatmap for the NLoS reflector scenario.

It identifies the optimal beam pair at $T_x = 27^\circ$, $R_x = -30^\circ$, in agreement with the previous search results.

5.4 Discussion on the Results

The results obtained across the three experimental scenarios highlight both the potential and the limitations of the evaluated beam search strategies. Several key observations and remarks can be drawn regarding the performance of the approaches and the practical constraints of the hardware.

Beamwidth Limitations and RSS Sensitivity

A primary limitation of the experimental setup stems from the relatively broad beamwidths achievable with the available antenna modules. The hardware does not allow the generation of narrow pencil beams, which restricts the resolution of the algorithms and their ability to discriminate between adjacent candidate directions. This results in situations where several neighboring beams produce nearly identical RSS values, complicating the decision-making process and reducing angular precision.

On the one hand, this limitation hinders fine-grained alignment, especially in the later refinement stages where adjacent beams are expected to capture subtle variations in received power. On the other hand, the use of wide beams is not without benefit.

Broader radiation patterns allow for a more diffuse distribution of power over the angular domain, which can be advantageous during the early stages of initial access. Even though counterintuitive when precise angular alignment is the ultimate goal, wide beams increase the robustness of the initial detection process. A trade-off therefore arises between angular precision and sufficient power to establish the initial link. Introducing a threshold based on the minimum required received power needed at the receiver could provide an effective mechanism to arbitrate between these two conflicting objectives.

Sensitivity to Physical Conditions

Another important factor influencing the results is the high sensitivity of the system to the physical arrangement of the antennas. Minor variations in the relative placement of Tx and Rx, as well as inaccuracies in the manual pointing of the antenna panels, may introduce noticeable angular deviations. These uncertainties limit the precision of the experimental outcomes and must be acknowledged when interpreting the absolute values of the reported deviations.

Intra-Sector Refinement Challenges

The hierarchical search results also indicate that the refinement phases (P2 and P3) are particularly affected by the beamwidth constraints. In these stages, the task is to discriminate between candidate beams located within a confined angular sector. However, the relatively thick beams achievable with the hardware often yield RSS values that are too similar to enable reliable discrimination. As a result, the RSS metric alone is not always sufficient to clearly identify the optimal beam. This limitation underscores the need for finer beamforming capabilities or the need to introduce a different indicator to discriminate between adjacent candidate beam pairs, particularly in the critical intra-sector refinement phases where angular precision is expected to play a decisive role.

Global Perspective

From a broader perspective, the experiments indicate that exhaustive scans, while theoretically capable of achieving the most precise alignment, require a disproportionately large allocation of resources in both time and signaling overhead. The performance gain observed does not necessarily justify such costly procedures, especially in the context of initial access where efficiency is important.

Conversely, hierarchical strategies consistently converged towards beam pairs of comparable quality, achieving satisfactory SNR levels despite the lower gain of wide beams used in the early phases. Although wide beams introduce increased risk of misdetection and reduce angular resolution, the algorithm is able to compensate in subsequent refinement steps. The final beam pairs obtained still provide reliable alignment and strong signal quality, confirming the practical suitability of hierarchical search strategies for real-world deployment.

6 Conclusion

The exploration of beam alignment strategies in the context of high-frequency mobile communications, particularly within millimeter wave systems, has highlighted several insights. The inherent directionality of mmWave propagation has deeply influenced the design of the 5G standard itself, where initial access procedures rely on efficient and low-cost mechanisms at the user equipment side. To this end, the standard defines a set of reference signals that allow devices to perform detection and establish initial connectivity. While the signals themselves are formally specified, the actual detection and alignment mechanisms are left as an open design choice, enabling a range of strategies where beamforming plays a central role.

The targeted review of initial access methods revealed a trend in the literature toward solutions that prioritize rapid access, minimal computational, and efficient extraction of environmental information, while remaining compatible with the hardware limitations encountered in UEs. Approaches explored include exhaustive and hierarchical search schemes, compressed sensing-based methods, and more recently, machine learning-driven solutions. Among these, hierarchical strategies stand out as a pragmatic balance, combining reduced search effort with robustness, and aligning well with the iterative procedures implicitly favored by the standard.

Within this thesis, a deliberate choice was made to implement and experimentally evaluate solutions based on sequential angular scanning. Despite limitations related to antenna beamwidth and the constraints of the experimental testbed, the results demonstrated meaningful insights into the trade-offs between accuracy and algorithmic complexity. In particular, the hierarchical method showed a clear advantage by achieving competitive performance at a fraction of the cost associated with exhaustive search. This tendency resonates with the broader orientation of the standard and with the persistent interest in low-complexity methods observed across the literature.

Overall, despite the limited guidance provided by the 3GPP specifications regarding concrete implementation strategies for initial beam alignment, the objective of this work, aiming to identify, implement, and experimentally validate a representative set of candidate solutions, was realized and yielded expected insights towards the aforementioned low complexity strategies. The results obtained justify the adoption of such schemes, especially in practical scenarios where multiple antenna panels are deployed at the UE side. Maintaining a reasonable complexity is thus essential to ensure low latency and manageable processing overhead, making scalable lightweight approaches not only attractive but necessary.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work the author(s) used ChatGPT in order to generate illustration materials, perform grammar correction, provide translations, suggest rephrasing, understand scientific materials, and search for relevant literature references. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Appendices

All source code implementations employed in this work are publicly accessible via the following GitHub repository.

Repository hyperlink : [Repository].

Url Access : <https://github.com/Scarluzz/Environment-Mapping-with-28-GHz-BeamSteering-Transceivers>